OS/7

Administrator Manual

Version 1.0.0 (preview)

Ubuntu 26.04 resolute

Snapshot 20260909T000000Z

PowerShell 7.6.5

build/config/os7-release.conf

Contents

1 Introduction

1.1 What OS/7 is

OS/7 is a Linux operating system built on **Ubuntu 26.04 LTS** for administrators who come from the Microsoft world. It does not change what Linux is — it changes what you drive it with and how you manage it.

Four decisions shape everything that follows:

PowerShell is the interface. Log in and you land in PowerShell, not `bash`. The administrative commands are cmdlets with the familiar verb-noun names, and they return **objects** rather than lines of text — `Get-OS7Service -Detailed | ? { $_.Healthy -eq $false }` is a sensible line and needs no `grep`, no `awk` and no assumption about column widths.

Management goes through Microsoft's own tools. Entra ID for sign-in, Intune for device management, Azure Arc for inventory. Where a technical preference of OS/7's collides with something those services require, the service wins — which matters most for the disk layout, the encryption, and the names the machine reports itself under.

The filesystem is ZFS, and updates roll back. An update never modifies the running system. It makes a copy, changes the copy, and switches afterwards. If something goes wrong, the way back is one command and a restart — not a restore from backup.

Setup is a text-mode installer styled after MS-DOS 6.22 and the Windows 2000 text phase. That is not nostalgia: an installer that works on any serial console and in any remote-management window is worth more on a server than one that needs a graphics stack.

1.2 Who this manual is for

Administrators who run OS/7 machines. It assumes you are comfortable with Windows Server and PowerShell. Linux knowledge helps but is nowhere assumed: where a Linux concept is unavoidable — ZFS, systemd, netplan — the manual explains it at the point where it is needed.

The manual is organised **by task**, not by module. If you want to know how an update is rolled back, that is chapter 5, with everything that belongs to it. The complete list of every command is **Appendix A**.

1.3 The two architectures

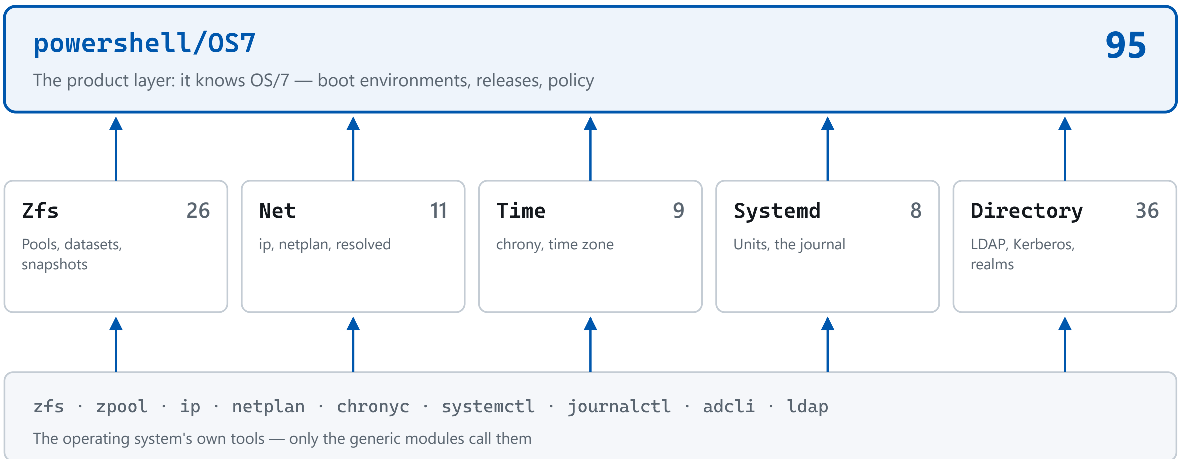
OS/7 exists for two processor architectures, and they are **not the same product**:

	amd64 (x86-64)	arm64 (AArch64)
Desktop	yes — GNOME in the classic appearance	no
Microsoft Edge	yes	no
Intune portal	yes	no
Server / headless	yes	yes
PowerShell, ZFS, boot environments, updates, network, time, directory	yes	yes

The reason for the split is simple: Microsoft ships no desktop stack for arm64. Everything OS/7 builds itself exists on both; everything that comes from Microsoft exists where Microsoft ships it. Installing on amd64 you choose between **GUI** and **Headless**; on arm64 the question does not arise.

1.4 How the system is put together

The commands in this manual come from six PowerShell modules. The split is not cosmetic — it is the reason the commands behave alike:



The product layer may not reach a subsystem directly. Checked, not asserted:
`installer/testing/check-layering.py`

> The six modules and the direction between them. The five generic modules each know one subsystem of the operating system and nothing about OS/7; the OS7 module is the product on top of them and holds every decision that is OS/7-specific.

What that means in practice:

- `Get-Zpool` **works on any Ubuntu.** The generic modules are tools for their subsystem and nothing else. They know no boot environments, no releases and no policy.
- `Get-OS7BootEnvironment` **exists only here.** Anything with `OS7` in the name is product knowledge and rests on the layer below it.
- **The `OS7` prefix is the canonical surface** — and the Windows names are there as well. `Get-OS7Service` is the cmdlet this manual teaches: its parameters are systemd's, and `Healthy` answers

four questions rather than one. Beside it, though, you will also find `Get-Service`, `Start-Service`, `Set-TimeZone`, `Get-ComputerInfo`, `Rename-Computer` and eight more names out of Microsoft's `Microsoft.PowerShell.Management` that PowerShell does not ship on Linux. They carry **Windows' parameters and Windows' output columns**, so that a script copied off a Windows machine runs here instead of stopping on its first line.

Anything such a cmdlet cannot do on this platform it **refuses by name**, saying what to use instead — it never ignores the parameter quietly. Appendix C lists all fourteen.

1.5 Two answers, never one

One pattern runs through the whole command surface and is unfamiliar at first reading: **where there is a stored intent and a running reality, OS/7 reports both, separately.**

`Get-OS7NetworkConfiguration` tells you what netplan says **and** what is actually on the adapters. `Get-OS7Service` tells you what systemd thinks of the unit **and** whether the service is genuinely well. The interesting machine is the one where the two disagree — a merged field would hide exactly that machine.

For the same reason every cmdlet asks the thing itself rather than trusting a tool's report: `netplan apply` returns 0 for a configuration that brought nothing up; `systemctl is-active` says `active` for a service in a restart loop; `useradd -m` exits 0 and creates no home when the directory already exists. An exit code is a diagnostic, not an answer.

1.6 Conventions in this manual

Input is shown in code blocks, exactly as it is typed:

```
Get-OS7BootEnvironment | Format-Table Name, Active, Running
```

Screen illustrations show the console of a running OS/7 machine.

> **Elevation.** Nearly every changing cmdlet and many reading ones need administrative rights, because they touch ZFS, systemd or files under `/etc`. On OS/7 you start an elevated PowerShell for that: `powershell sudo pwsh` Every example in this manual assumes you are in such a session unless it says otherwise.

2 Getting started

2.1 Logging in

An OS/7 machine without a desktop greets you at the console with a sign-in prompt that names the machine's version. After you log in a greeting appears, and then **a PowerShell prompt directly**.

```
[  OK  ] Started systemd-timedated.service - Time & Date Service.
[  OK  ] Finished snapd.seeded.service - Wait until snapd is fully seeded.
[  OK  ] Finished systemd-networkd-wait-onl…ce - Wait for Network to be Online.
[  OK  ] Reached target network-online.target - Network is Online.
[  OK  ] Started update-notifier-download.t…hat failed at package install time.
[  OK  ] Started update-notifier-motd.timer… a new version of Ubuntu available.
[  OK  ] Reached target timers.target - Timer Units.
[  OK  ] Started cups-browsed.service - Mak…te CUPS printers available locally.
[  OK  ] Reached target multi-user.target - Multi-User System.
         Starting authd.service - authd daemon service...
[  OK  ] Finished sanoid.service - Snapshot ZFS filesystems.
         Starting sanoid-prune.service - Prune ZFS snapshots...
[  OK  ] Started authd.service - authd daemon service.
[  OK  ] Finished sanoid-prune.service - Prune ZFS snapshots.

OS/7 1.0.0 (preview) os7-s5 ttyS0

os7-s5 login: os7admin
Password:
OS/7 1.0.0 (preview)
kernel 7.0.0-30-generic (x86_64), boot environment os7_1.0.0.175_202609030759


PS /home/os7admin>
```

> The console after logging in: /etc/issue names the version, the greeting follows, and the login lands in PowerShell — not at a bash prompt.

That holds for all three ways onto the machine:

Route	Where you land
Console (screen or serial line)	PowerShell
`ssh user@machine` (interactive)	PowerShell
`ssh user@machine 'command'`	**bash**
A terminal window on the desktop	PowerShell

The third line matters and is deliberate. A non-interactive SSH invocation stays in `bash`, because `scp`, `rsync`, `git` and every tool that fires a command over SSH depends on it. Switching that path too would not produce a PowerShell prompt — it would break every file transfer to the machine.

2.2 Which machine is this?

The first question on an unfamiliar machine:

```
Get-OS7Version
```

```
OS/7 1.0.0 (preview)
Ubuntu 26.04 base, Server, amd64


PS /home/os7admin>
```

> Get-OS7Version names the version. The output is an object, not a line of text.

Every field the machine knows about itself — the commit it was built from, the archive snapshot it was built against, the boot environment it is running out of:

```
Get-OS7Version | Format-List *
```

```
ProductName  : OS/7
Known        : True
Version      : 1.0.0
Build        : 175
FullVersion  : 1.0.0.175
Channel      : preview
Edition      : Server
Architecture : amd64
BaseRelease  : 26.04
Built        : 09/02/2026 15:34:27
Reproducible : True
Short        : 1.0.0 (preview)
Full         : 1.0.0.175 (preview)
Display      : OS/7 1.0.0 (preview)
Drift        :
ManifestPath : /usr/lib/os7/release.json


PS /home/os7admin>
```

> All the version fields. They come from /usr/lib/os7/release.json, the same file Setup and the boot-menu entry read.

Two ways of writing a version

OS/7 shows a **person** three fields and a **machine** four:

	Example	Where it appears
short	`1.0.0 (development)`	title row, `PRETTY_NAME`, `/etc/issue`, the greeting, `Get-OS7Version`
long	`1.0.0.159`	dataset names, `IMAGE_VERSION`, the ISO filename, `--version`, the boot menu, Setup's Complete screen

This is a **display rule, not a second number**. Wherever two builds have to be told apart, the long form appears; wherever a human reads, the short one.

What the machine calls itself, and to whom

An OS/7 machine says **OS/7 where a person looks, and Ubuntu where software looks.** Branded: `PRETTY_NAME`, `/etc/issue`, the greeting header and the GRUB menu. Not branded: `NAME`, `ID`, `ID_LIKE`, `VERSION`, `VERSION_ID`, `VERSION_CODENAME`, `/etc/lsb-release` and `uname`.

The reason is practical: Microsoft's onboarding script for Azure Arc reads `NAME` and gives up if it does not contain `buntu`. Other tools read other fields. So every branded surface reads a file of OS/7's own (`/usr/lib/os7/product`) and none of the `os-release` fields.

2.3 Finding the commands

Every administrative command lives in six modules present on every OS/7 machine:

```
Get-Module -ListAvailable OS7,Zfs,Net,Time,Systemd
```

```
    Directory: /usr/local/share/powershell/Modules

ModuleType Version    PreRelease Name                                PSEdition
---------- -------    ---------- ----                                ---------
Script     1.0.0.175             Net                                 Desk
Script     1.0.0.175             OS7                                 Desk
Script     1.0.0.175             Systemd                             Desk
Script     1.0.0.175             Time                                Desk
Script     1.0.0.175             Zfs                                 Desk


PS /home/os7admin>
```

The modules as the machine reports them, with their paths and versions.

From here you work with PowerShell's own tools. Everything on one topic:

```
Get-Command -Module OS7 -Noun OS7BootEnvironment
```

```
CommandType     Name                                          Version    S
                                                                         o
                                                                         u
                                                                         r
                                                                         c
                                                                         e
-----------     ----                                          -------    -
Function        Get-OS7BootEnvironment                        1.0.0.175  O
Function        New-OS7BootEnvironment                        1.0.0.175  O
Function        Remove-OS7BootEnvironment                     1.0.0.175  O
Function        Set-OS7BootEnvironment                        1.0.0.175  O

PS /home/os7admin>
```

> Every command about boot environments, found through the noun.

Wildcards widen the topic:

```
Get-Command -Module OS7 -Verb Get -Noun OS7B*
```

```
CommandType     Name                                          Version    S
                                                                         o
                                                                         u
                                                                         r
                                                                         c
                                                                         e
-----------     ----                                          -------    -
Function        Get-OS7BackupCoverage                         1.0.0.175  O
Function        Get-OS7BackupPolicy                           1.0.0.175  O
Function        Get-OS7BackupStatus                           1.0.0.175  O
Function        Get-OS7BackupTarget                           1.0.0.175  O
Function        Get-OS7BootEnvironment                        1.0.0.175  O

PS /home/os7admin>
```

> The Get commands whose noun starts with OS7B: boot environments and backup.

And onwards:

```
Get-Command -Module OS7 -Noun *Backup*     # everything about backup
Get-Command -Module OS7 -Verb Set          # everything that changes something
Get-Help Update-OS7 -Full                  # the full help
Get-Help Restore-OS7 -Examples             # just the examples
```

The complete list of all 194 commands is **Appendix A**.

2.4 The naming rules

Knowing the rules means looking things up less often.

A `Get-` never changes anything. Without exception.

A `Test-` answers a question and, on "no", says which part failed. `Test-OS7Network` does not merely report that an endpoint is unreachable; it says which one, and at which step the chain broke.

A `Set-` checks afterwards. `Set-OS7NetworkAdapter` writes the configuration, applies it, asks the kernel whether an address is actually present — **and puts the old configuration back when it is not.**

Destructive commands ask. `Remove-OS7BootEnvironment`, `Remove-ZfsDataset`, `Remove-Zpool` and `Restore-ZfsSnapshot` are classed as high-impact and require confirmation. In a script you suppress it with `-Confirm:$false`, and then it is a deliberate decision.

Secrets are never strings. Every cmdlet that takes one takes a `[securestring]` or a `[pscredential]`. There is no parameter that accepts a password in clear text, and none of these objects reaches an output that can be serialised.

2.5 Where to start

On an unfamiliar machine these five commands give a complete picture in about a minute:

```
Get-OS7Version                  # which OS/7 is this
Get-OS7BootEnvironment          # what can it boot, what is running
Get-OS7Service -Detailed | ? { $_.Healthy -eq $false }   # what is not well
Test-OS7Network                 # can it get out, and to where
Get-OS7ManagementStatus         # is it managed, and if not, why not
```

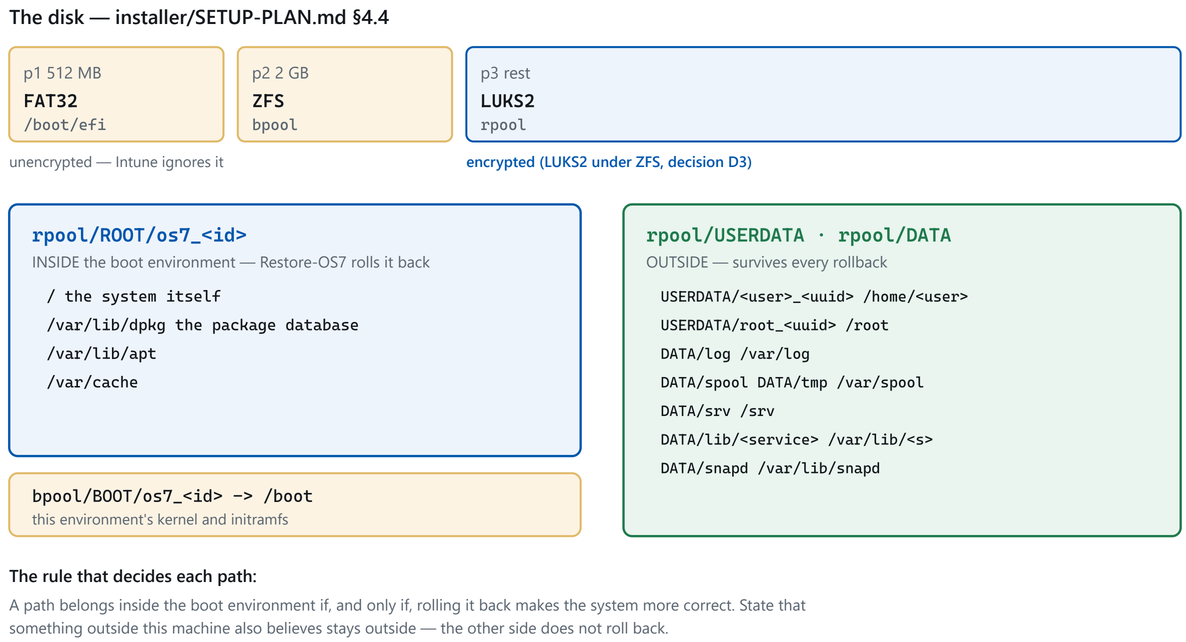
Chapter 14 builds a complete diagnostic order out of these.

3 Installing a machine

Installation is done with `os7-setup`, OS/7's text-mode installer. It starts from the installation medium by itself and needs no graphics card, no mouse and no network.

3.1 What Setup lays down

Before you press a key it is worth knowing what ends up on the disk — because this layout is the reason updates roll back later:



> The disk after installation. What matters is what lies inside and what lies outside the boot environment.

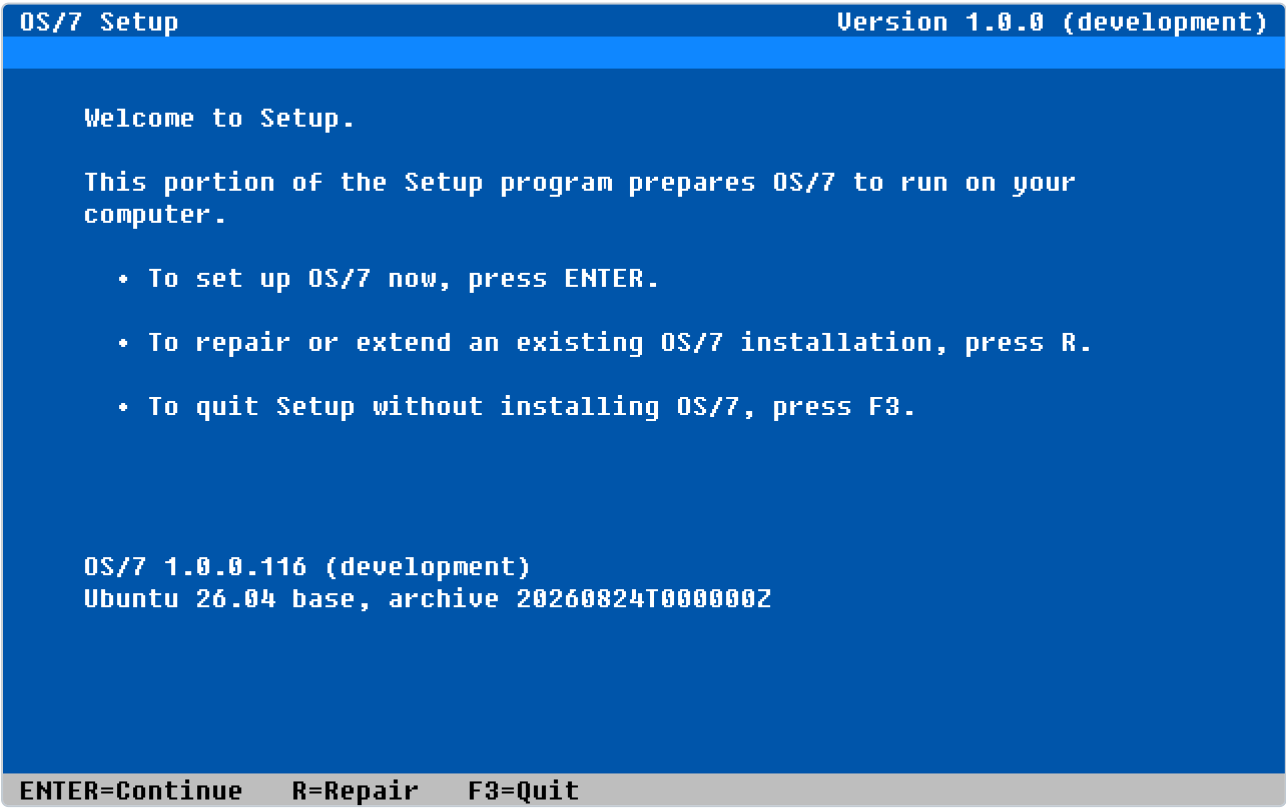
Three partitions:

- an **EFI System Partition** (512 MB, FAT32) — unencrypted, because the firmware must be able to read it;
- a **boot pool** `bpool` (2 GB, ZFS) — unencrypted, because GRUB must be able to read it, and with a restricted feature set GRUB understands;
- the rest as a **LUKS2 container**, with the ZFS pool `rpool` inside it.

Encryption is **LUKS2 under ZFS**, not ZFS-native encryption. That is a deliberate decision: LUKS is what Microsoft's management tooling on Linux recognises and reports as disk encryption.

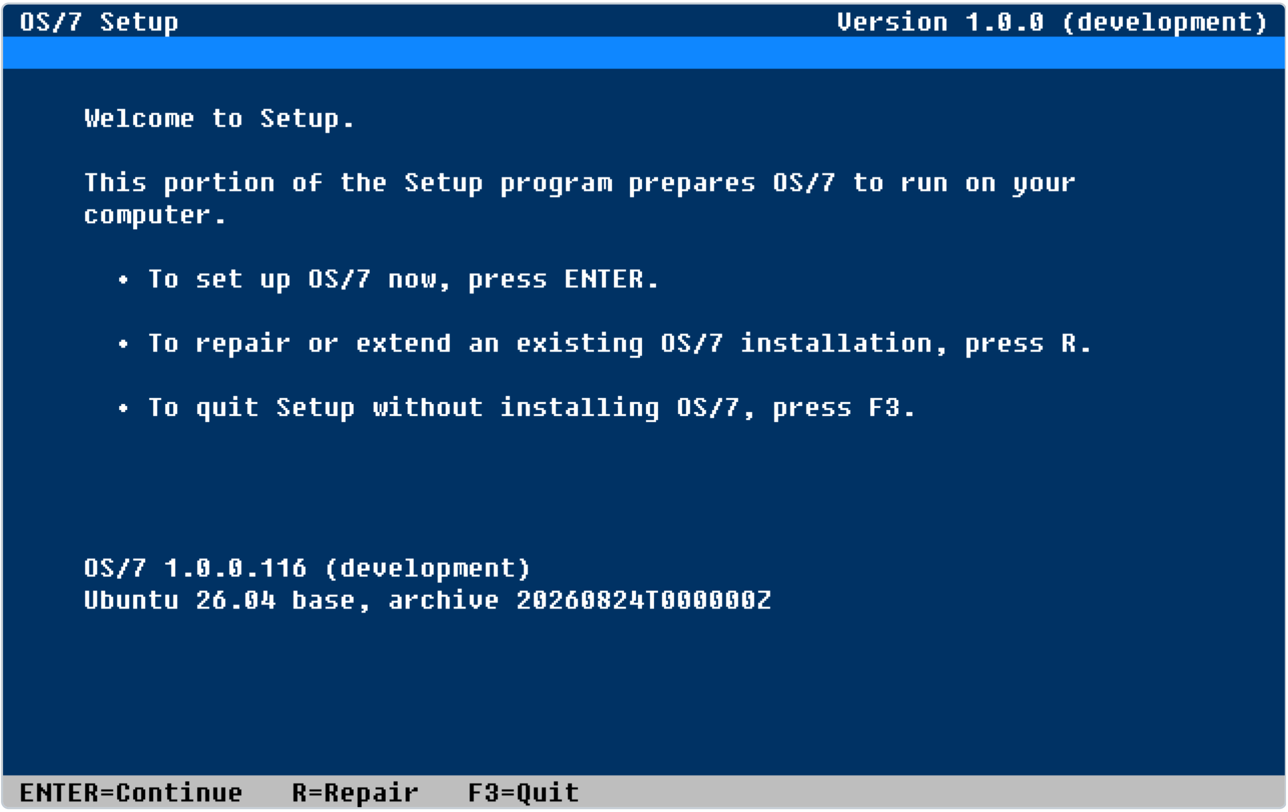
3.2 The screens, in order

1 — Welcome



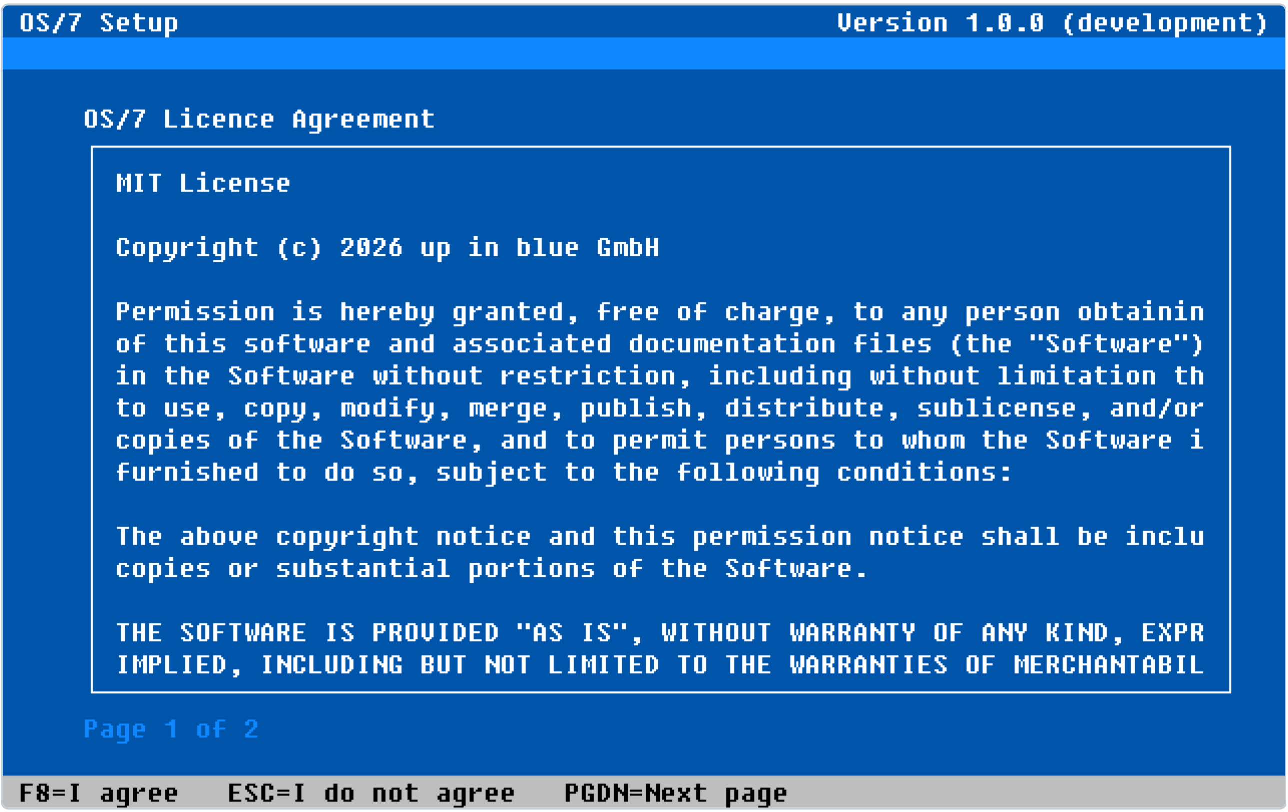
The first screen. The version sits at the top right of every screen.

`ENTER` installs, `R` repairs an existing installation, `F3` quits. `F5` switches Setup to a higher-contrast field — for remote-management windows and poor displays.



The same screen with F5: a darker field, the same controls.

2 — Licence



The licence terms, a page at a time with PAGE UP and PAGE DOWN.

`F8` accepts, `ESC` declines.

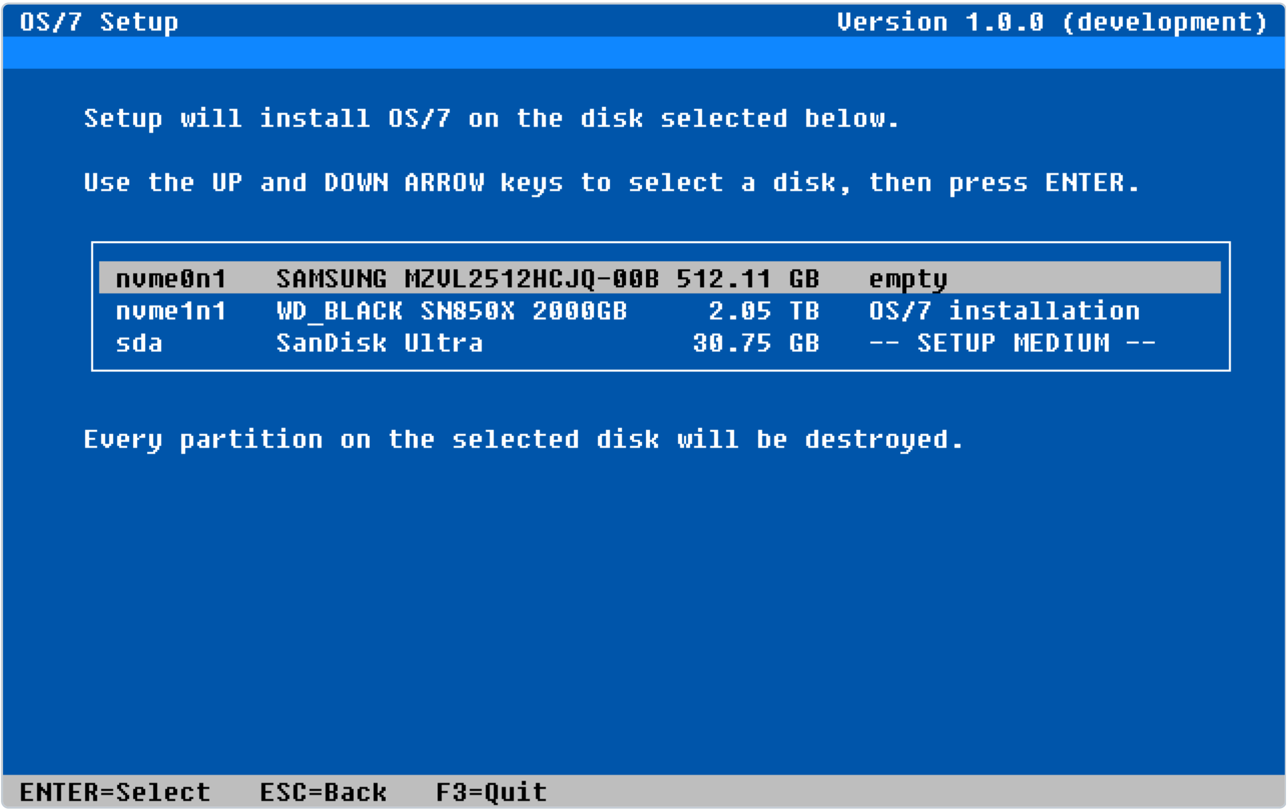
3 — Regional settings

```
OS/7 Setup                                    Version 1.0.0 (development)

    Setup will use the following regional settings:

      Language:           English (United States)  (en_US.UTF-8)
      Keyboard:           English (US)  (us)
      Time zone:          UTC

      The settings are correct.

    If all the settings are correct, press ENTER.

    To change a setting, press the UP or DOWN ARROW keys to select it.
    Then press ENTER to see alternatives.

ENTER=Continue   ↑↓=Select   F3=Quit
```

Language, keyboard layout and time zone on one screen.

What you choose here sets the language of the installed system, the console keyboard layout and the time zone. The zone can be changed later with `Set-OS7TimeZone`.

4 — Select a disk



The machine's disks. The installation medium is listed and not selectable.

Setup lists every disk and says what is on each. A disk that already carries an OS/7 installation is recognised as such. The installation medium itself appears in the list but is blocked — visible, so that nobody wonders where it went.

5 — Layout and encryption

```
OS/7 Setup                                          Version 1.0.0 (development)

     Setup will use the following storage settings:

        Disk:                  nvme0n1  (512.11 GB)
        Layout:                single disk
        EFI partition:         512 MB   FAT32
        Boot pool:             2 GB     ZFS  (bpool, GRUB-readable features)
        Root pool:             509.43 GB   ZFS  (rpool)
        Encryption:            LUKS2, aes-xts-plain64  (passphrase)
        Passphrase:            -- not set --
        Swap:                  zram, 50% of RAM (no swap on disk)

        The settings are correct.

     If all the settings are correct, press ENTER.

     To change a setting, press the UP or DOWN ARROW keys to select it.

ENTER=Continue   ↑↓=Select   ESC=Back   F3=Quit
```

The layout at a glance, with encryption and swap.

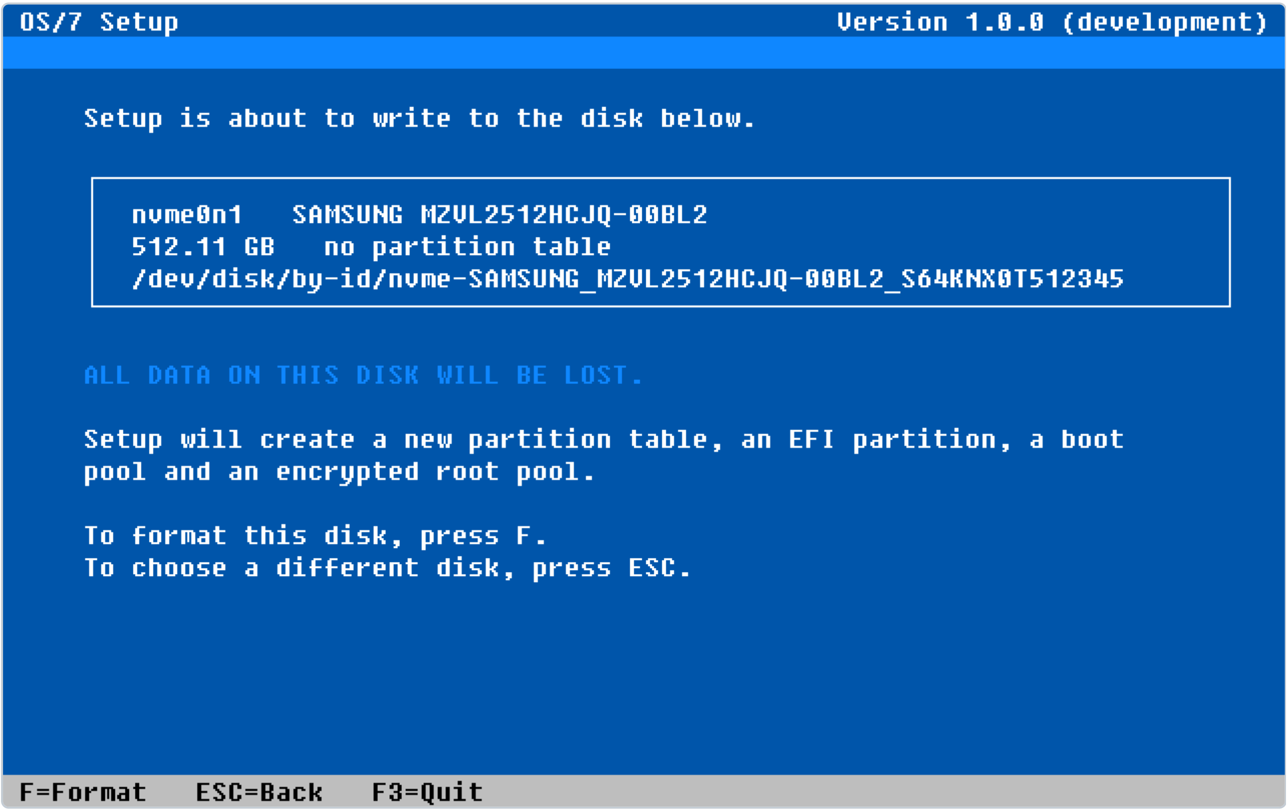
This is where the disk-encryption passphrase is set. Until it is, the line reads `-- not set --` and Setup will not continue.

```
OS/7 Setup                                        Version 1.0.0 (development)

    Setup will use the following storage settings:

     ┌────────────────────────────────────────────────────────────────────┐
     │  Disk:              nvme0n1  (512.11 GB)                           │
     │  Layout:            single disk                                    │
     │  EFI partition:     512 MB   FAT32                                 │
     │  Boot pool:         2 GB     ZFS  (bpool, GRUB-readable features)  │
     │  Root pool:         509.43 GB   ZFS  (rpool)                       │
     │  Encryption:        LUKS2, aes-xts-plain64  (passphrase)           │
     │  Passphrase:        set                                            │
     │  Swap:              zram, 50% of RAM (no swap on disk)             │
     ├────────────────────────────────────────────────────────────────────┤
     │  The settings are correct.                                         │

    If all the settings are correct, press ENTER.

    To change a setting, press the UP or DOWN ARROW keys to select it.

ENTER=Continue   ↑↓=Select   ESC=Back   F3=Quit
```

Once the passphrase is set the screen is complete.

Swap defaults to **zram** — compressed, in memory — and not to the disk. A swap file on ZFS is not an option; the combination deadlocks.

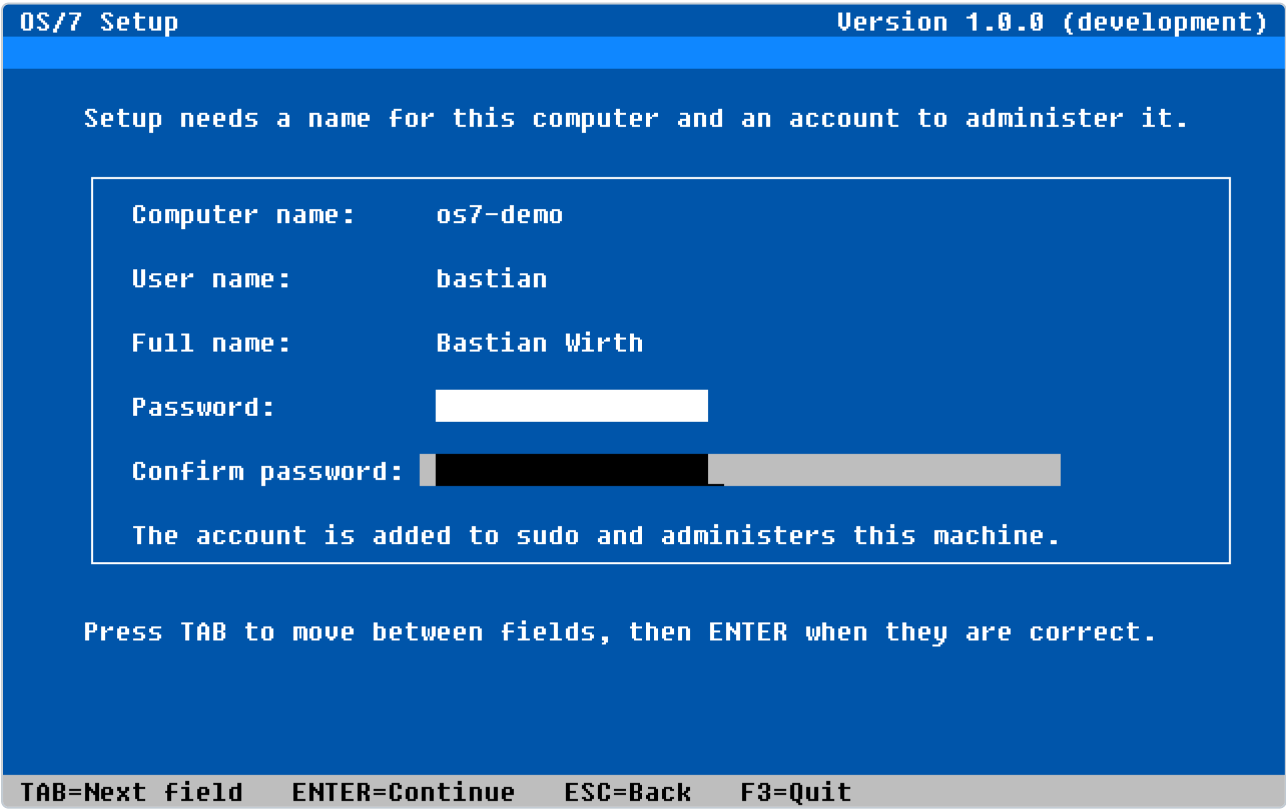
6 — Confirmation



The last screen before anything is written.

This is the gate. Up to here **nothing** has been written to the disk. `F` confirms, `ESC` goes back. Writing starts at screen 10 — screens 7 to 9 are filled in while the disk is still untouched.

7 — Computer name and administrator account



Computer name, login name, full name and password.

The account created here is the machine's first administrator account. Its home directory gets a ZFS dataset of its own under `rpool/USERDATA` — outside the boot environment, so that a later rollback does not take the user's files with it.

8 — Install mode (amd64 only)

```
OS/7 Setup                                          Version 1.0.0 (development)

    Choose how this computer will be used.

    Use the UP and DOWN ARROW keys to select, then press ENTER.

      Desktop  - GNOME, and eligible for Microsoft Intune enrolment
      Headless - no desktop; managed through Azure Arc

    Desktop installs GNOME from this medium. It is the only mode
    Microsoft Intune can enrol, because supported Linux enrolment
    requires a GNOME session.

    Headless removes the desktop. Nothing is downloaded either way -
    both modes come from the medium Setup booted from.

ENTER=Continue   ESC=Back   F3=Quit
```

The choice between desktop and headless.

GUI installs the classic OS/7 desktop with Microsoft Edge and the Intune portal. **Headless** installs a server with no graphics stack.

```
OS/7 Setup                                          Version 1.0.0 (development)

    Choose how this computer will be used.

    Use the UP and DOWN ARROW keys to select, then press ENTER.

      Desktop  - GNOME, and eligible for Microsoft Intune enrolment
      Headless - no desktop; managed through Azure Arc

    Desktop installs GNOME from this medium. It is the only mode
    Microsoft Intune can enrol, because supported Linux enrolment
    requires a GNOME session.

    Headless removes the desktop. Nothing is downloaded either way -
    both modes come from the medium Setup booted from.

ENTER=Continue   ESC=Back   F3=Quit
```

The same choice with the headless entry selected.

On arm64 this screen does not appear, because there is only headless.

9 — Network

```
OS/7 Setup                                          Version 1.0.0 (development)

    Setup can configure this computer's network connection now.

      Adapter
      enp0s31f6     Ethernet    e1000e          link up
      wlp3s0        Wi-Fi       iwlwifi

      Method
      Obtain an address automatically (DHCP)
      Specify an address (static TCP/IP)
      Leave this computer without a network connection

    Setup will apply these settings now and test them before
    writing them to the installed system. Press T to test.

    Not yet tested.

TAB=Next list   F4=Test   ENTER=Continue   ESC=Back   F3=Quit
```

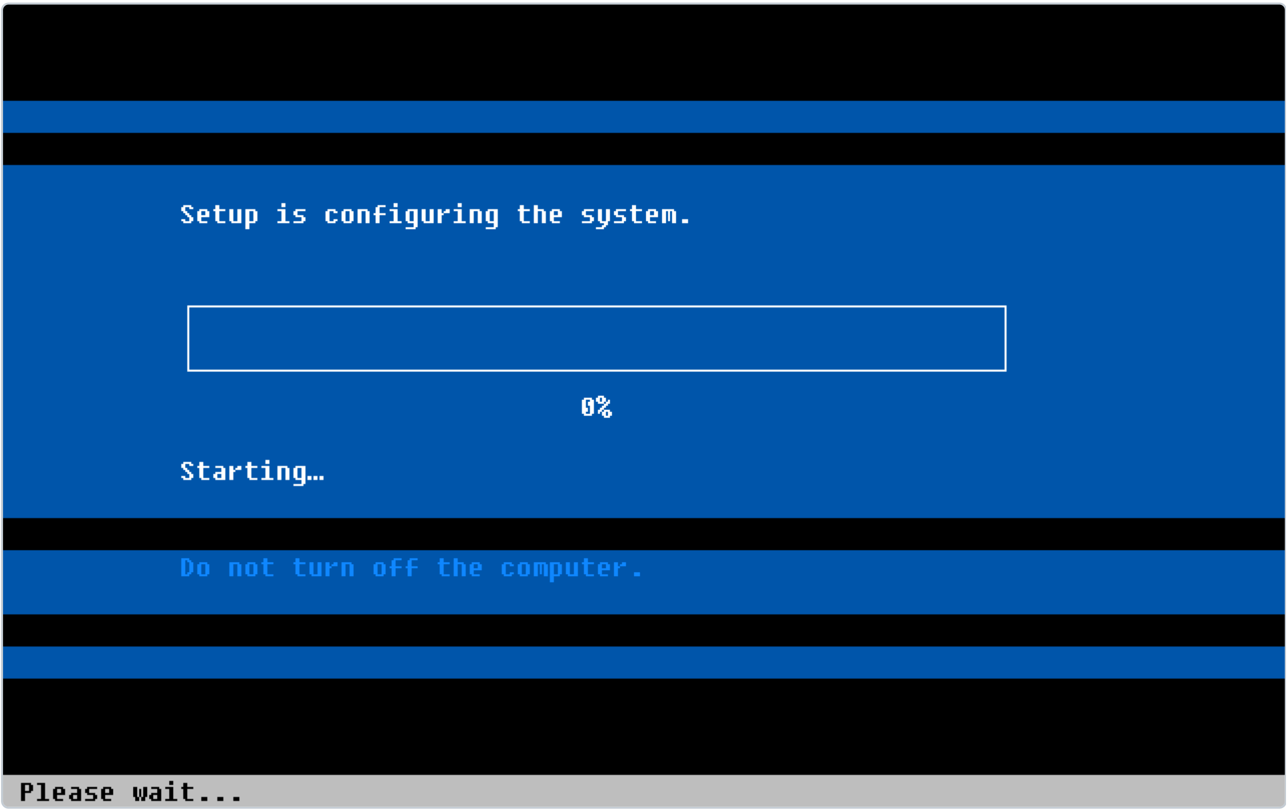
Adapter and method. Setup applies the settings and tests them before writing them to the installed system.

Setup shows the adapters that actually exist, with kind, driver and link state. Three methods: DHCP, a static address, or no network configuration at all.

The line at the bottom is the notable one: `F4` **tests the setting before it is committed.** A typed static address that leads nowhere shows up here rather than after the first restart.

A static address leads to screen **9S** with address, prefix length, default gateway, name servers and search domains; a wireless adapter leads to screen **9W** with network name and authentication.

10 and 11 — Copying and configuring



The progress display while the disk is written.

From here Setup works: partition, create LUKS, create the pools, lay down the datasets, unpack the filesystem, configure the system, install the bootloader, create the account and prepare the TPM unlock.

Every step proves itself — it does not note that a program returned 0, it asks for the result. The record of those proofs lands on the installed machine at `/var/log/os7-setup/install.log` and is readable later with `Get-OS7InstallLog`.

12 — Complete

```
OS/7 Setup                                              Version 1.0.0 (development)

    Setup has prepared this computer as follows.

       OS/7 version: 1.0.0.116 (development)
       Language:     en_US.UTF-8
       Keyboard:     us
       Time zone:    UTC
       Disk:         /dev/disk/by-id/nvme-SAMSUNG_MZVL2512HCJQ-00BL2_S64KNX0T51
       Encryption:   LUKS2 (passphrase set)
       Swap:         zram
       Computer:     os7-demo
       Account:      bastian   (desktop)
       Network:      enp0s31f6  DHCP

    Remove the setup medium and press ENTER to restart.
    This computer will ask for the disk passphrase when it starts.
    The network settings were NOT tested before they were written.
    A record of this installation is at /var/log/os7-setup/install.log.

ENTER=Restart   F3=Quit to a shell
```

The Complete screen names the full version of the installed machine.

3.3 Installing unattended

Setup takes a complete installation plan as a file and then runs without input:

```
os7-setup --unattend /path/to/plan.json
```

The plan is JSON and holds exactly what screens 3 to 9 collect:

```json
{
  "version": 1,
  "intent": "Install",
  "language": "en_GB.UTF-8",
  "keyboard": "gb",
  "timezone": "Europe/London",
  "mode": "Headless",
  "storage": {
    "disk": "/dev/disk/by-id/nvme-...",
    "layout": "single",
    "efiMiB": 512,
    "bpoolGiB": 2,
    "encrypt": true,
    "swap": "zram"
  },
  "account": {
    "hostname": "os7-srv-01",
    "username": "os7admin",
    "fullName": "OS/7 Administrator"
  },
  "network": {
    "interface": "auto",
    "kind": "Wired",
    "method": "Dhcp"
  }
}
```

The secrets are **not** in the plan. The passphrase and the password are passed separately, so a plan can safely sit in a deployment share.

The whole plan is validated **once**, immediately before the first write. That is the only place where "nothing downstream will catch this" is a true sentence.

3.4 Checking Setup before trusting it

Setup carries a self-test:

```
os7-setup --self-test
```

It checks the palette, the font, glyph coverage and key decoding. When something about an installation looks wrong, this is the first command — it runs in seconds and needs no disk.

3.5 After installation

Two things happen automatically on the first boot: the **TPM2 unlock** is enrolled against the installed machine's actual boot chain, and the release's **first-boot migrations** run.

The TPM enrolment belongs on the first boot rather than in the installer for a precise reason: the installation medium boots differently from the installed machine, and a key sealed against the medium's state will not open on the finished machine. The result is visible on the second boot — it no longer asks for the passphrase.

After that, log in and check the machine the way chapter 2.5 describes.

4 Storage and the filesystem

OS/7's filesystem is **ZFS**. Coming from Windows, think of it as disk management, the filesystem, shadow copies and software RAID combined — except that one layer manages all of it and all of it can be checked for consistency at any time.

Two terms are enough to start:

- A **pool** is the disks taken together. OS/7 creates two: `bpool` for what GRUB has to read, and `rpool` for everything else.
- A **dataset** is a filesystem inside a pool. Datasets share the pool's free space but have their own properties, their own snapshots and their own mount point. Where Windows would create a partition, ZFS creates a dataset — with no fixed size and no reformatting.

4.1 Looking at the pools

```
Get-Zpool | Format-Table Name,Health,Size,Free
```

```
Name  Health        Size        Free
----  ------        ----        ----
bpool ONLINE  2013265920  1888354304
rpool ONLINE 22548578304 17821884416

PS /home/os7admin>
```

The two pools of an OS/7 machine.

`Health` is the question that counts. `ONLINE` is good; anything else is a reason to look further:

```
Get-ZpoolStatus rpool | Format-List Name,State,Scan
```

```
Name  : rpool
State : ONLINE

PS /home/os7admin>
```

A pool's state, with its last scan.

`Get-ZpoolStatus` returns the vdev tree as **objects**, not as indented text. You can walk it instead of reading it:

```
(Get-ZpoolStatus rpool).Vdevs | Where-Object { $_.State -ne 'ONLINE' }
```

A full verification of the pool is started with:

```
Start-ZpoolScrub rpool
```

It runs in the background and barely affects service; progress appears in `Get-ZpoolStatus`'s `Scan` field.

4.2 The datasets

```
Get-ZfsDataset | Format-Table Name,Mountpoint
```

```
rpool                                                   /
rpool/DATA
rpool/DATA/lib
rpool/DATA/lib/authd                                    /var/lib/authd
rpool/DATA/lib/azcmagent                                /var/opt/azcmagent
rpool/DATA/lib/networkmanager                           /var/lib/NetworkManager
rpool/DATA/log                                          /var/log
rpool/DATA/snapd                                        /var/lib/snapd
rpool/DATA/spool                                        /var/spool
rpool/DATA/srv                                          /srv
rpool/DATA/tmp                                          /var/tmp
rpool/ROOT
rpool/ROOT/os7_1.0.0.175_202609030759                   /
rpool/ROOT/os7_1.0.0.175_202609030759/var               /var
rpool/ROOT/os7_1.0.0.175_202609030759/var/cache         /var/cache
rpool/ROOT/os7_1.0.0.175_202609030759/var/lib           /var/lib
rpool/ROOT/os7_1.0.0.175_202609030759/var/lib/apt       /var/lib/apt
rpool/ROOT/os7_1.0.0.175_202609030759/var/lib/dpkg      /var/lib/dpkg
rpool/USERDATA
rpool/USERDATA/os7admin_d5712bc8                        /home/os7admin
rpool/USERDATA/root_d5712bc8                            /root


PS /home/os7admin>
```

The dataset layout of an installed machine. The structure is the one from chapter 3.

This output is the layout chapter 3 drew:

What you see	What it is
`rpool/ROOT/os7_<version>_<time>`	the boot environment — the system itself
`rpool/ROOT/…/var/lib/dpkg`, `…/var/lib/apt`, `…/var/cache`	the package database, **inside** the boot environment
`rpool/DATA/log`, `…/spool`, `…/tmp`, `…/srv`, `…/lib/<service>`	state **outside** the boot environment
`rpool/USERDATA/<user>_<uuid>`	a home directory, outside the boot environment
`bpool/BOOT/os7_<version>_<time>`	this environment's kernel and initramfs

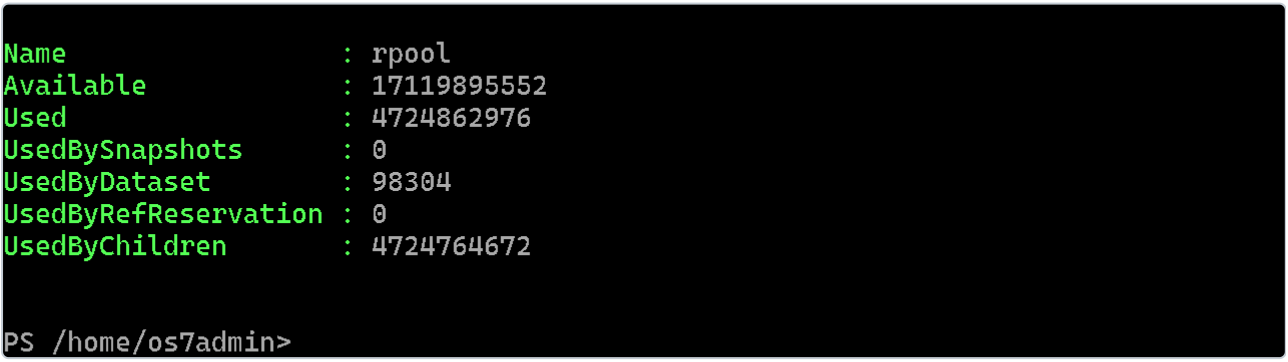
Sizes come back as `uint64` **bytes**, timestamps as `[datetime]`. You can calculate and sort without parsing:

```
Get-ZfsDataset | Sort-Object Used -Descending | Select-Object -First 5 Name,
    @{ n = 'Used'; e = { Format-ZfsSize $_.Used } }
```

4.3 Where the space went

ZFS answers "why is the pool full" differently from an ordinary filesystem, because snapshots hold space that no visible file needs any more:

```
Get-ZfsSpace rpool | Format-List
```



The breakdown of used space.

The fields separate what the data itself occupies, what **snapshots** are holding, and what is reserved for child datasets. A pool that is full with nothing visibly in it is holding its space in snapshots — and those come either from the backup policy (chapter 12) or from old boot environments (chapter 5).

4.4 Home directories

Every home directory should sit on a dataset of its own under `rpool/USERDATA`. This is not tidiness: a home **inside** the boot environment is taken back by a rollback along with the user's files, and no snapshot policy can follow it.

```
Get-OS7Home | Format-List
```

```
OS7-STEP run: stat -c %d /
OS7-STEP run: getent passwd os7admin
OS7-STEP run: stat -c %d /home/os7admin

Path          : /home/os7admin
Account       : os7admin
Uid           : 1000
Dataset       : rpool/USERDATA/os7admin_d5712bc8
OwnDataset    : True
OwnFilesystem : True
Agrees        : True
DeviceId      : 55
Note          : on its own dataset, outside the boot environment


PS /home/os7admin>
```

> For each home directory: does it have a dataset of its own, does it actually live there, and do those two answers agree.

The fields are chosen so that the fault shows, not only the intent:

Field	Question
`Dataset`	which dataset this home is meant to be on
`OwnDataset`	whether that dataset exists
`OwnFilesystem`	whether the home really is on a filesystem of its own
`Agrees`	whether those two say the same thing
`DeviceId`	the filesystem id that answer was derived from

`Agrees = False` is exactly the case the field exists to find: the dataset is there, and the home is somewhere else.

Moving a home onto its own dataset after the fact:

```
Move-OS7Home -UserName smith
```

The move creates the dataset, copies the content, verifies it and remounts. The original stays put at first; only `-RemoveOriginal` removes it, and only after the verification passed.

4.5 Snapshots by hand

The backup policy in chapter 12 takes snapshots on a schedule. Before a larger change one by hand is worth having:

```
New-ZfsSnapshot -Name rpool/USERDATA/smith_1a2b3c4d -SnapshotName before-migration
Get-ZfsSnapshot | Where-Object Name -like '*@before-migration'
```

A snapshot costs nothing at first and grows only by what changes afterwards. A single file is fetched back out of a snapshot with `Restore-OS7File` (chapter 12.7), without touching the dataset.

> **Careful.** `Restore-ZfsSnapshot` rolls an **entire dataset** back to the snapshot and discards everything newer, irreversibly. For "I only need this one file from yesterday", `Restore-OS7File` is the right tool.

4.6 What not to do

No swap on ZFS. Under memory pressure the combination deadlocks. OS/7 puts swap on zram, compressed in memory.

No snapshot policy on `rpool/ROOT` or `bpool/BOOT`. Those datasets belong to the boot-environment mechanism; a second thing taking and pruning snapshots there gets in its way. `Assert-OS7DatasetSafe` refuses it even if somebody tries.

Do not create datasets by hand where a cmdlet exists. `New-OS7Storage` and `New-OS7BootEnvironment` set properties a clone does not inherit — `canmount` and `mountpoint` in particular. A hand-made clone comes out `canmount=on mountpoint=none`: mountable over the running root, and invisible to every boot menu.

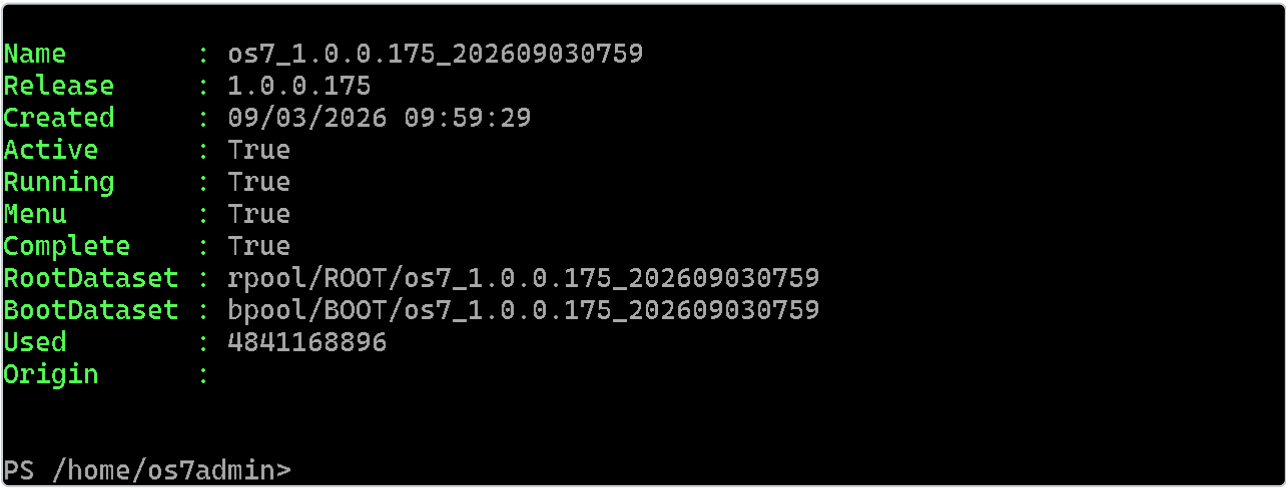
5 Boot environments and rollback

A **boot environment** is a complete, bootable system on the same disk. A machine can have several and always boots exactly one. It is the mechanism that turns an update into something that can be undone.

The idea is simple: instead of changing the running system, OS/7 makes a copy, changes the copy, and switches afterwards. Because ZFS is copy-on-write, the copy costs almost nothing at first — it grows only by the differences.

5.1 What is there

```
Get-OS7BootEnvironment
```



The machine's boot environments. The default output is one list per environment.

The fields each say something different, and the differences matter:

Field	Meaning
`Active`	ZFS has this environment mounted **somewhere**
`Running`	**the system is running out of this environment**
`Menu`	it has an entry in the boot menu
`Complete`	both halves — `rpool/ROOT` and `bpool/BOOT` — are present
`Release`	which OS/7 version is in it
`RootDataset`, `BootDataset`	the two datasets it consists of
`Origin`	the snapshot it was cloned from — empty for the installed one

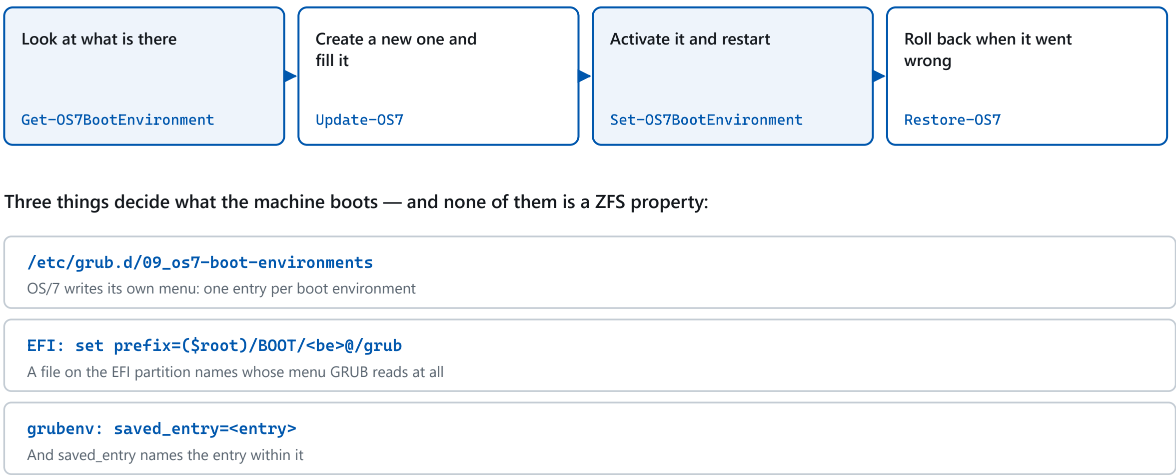
> **`Active` does not mean "running".** An update mounts the clone in order to fill it — at that moment two environments are `Active` and only one is `Running`. Everything that means "the running system" reads `Running`.

For an overview of several environments a table is handier:

```
Get-OS7BootEnvironment | Format-Table Name,Active,Running,Menu,Release
```

5.2 What decides what boots

Three things, and **none of them is a ZFS property**:



> The three places that together decide which boot environment starts.

1. **OS/7 writes its own boot menu** (`/etc/grub.d/09_os7-boot-environments`), with one entry per environment. The menu generators Ubuntu ships list exactly one environment per machine; a menu generated from one environment can never contain a second.
2. **A file on the EFI partition** decides whose menu GRUB reads at all (`set prefix=($root)'/BOOT/<environment>@/grub'`).
3. `saved_entry` in that environment's `grubenv` names the entry inside it.

In practice: switching is a **bootloader operation**, not a ZFS one. Activating a boot environment changes files on the EFI partition and in the boot pool — not a property on a dataset.

5.3 Creating a boot environment

Before a risky change — a configuration edit you may want to take back — create an environment without activating it:

```
New-OS7BootEnvironment -Name demo
```

```
OS7-STEP new boot environment demo from os7_1.0.0.175_202609030759
ZFS-STEP snapshot rpool/ROOT/os7_1.0.0.175_202609030759@demo
ParentContainsErrorRecordException: zfs snapshot -r rpool/ROOT/os7_1.0.0.175_202
609030759@demo exited 1 cannot
create snapshot 'rpool/ROOT/os7_1.0.0.175_202609030759@demo': dataset already
exists cannot create snapshot 'rpool/ROOT/os7_1.0.0.175_202609030759/var@demo':
dataset already exists cannot create snapshot
'rpool/ROOT/os7_1.0.0.175_202609030759/var/cache@demo': dataset already exists
cannot create snapshot 'rpool/ROOT/os7_1.0.0.175_202609030759/var/lib@demo':
dataset already exists cannot create snapshot
'rpool/ROOT/os7_1.0.0.175_202609030759/var/lib/apt@demo': dataset already
exists cannot create snapshot
'rpool/ROOT/os7_1.0.0.175_202609030759/var/lib/dpkg@demo': dataset already
exists no snapshots were created

PS /home/os7admin>
```

> Creating one shows every step: snapshots of both halves first, then the clones, then the result.

The output is deliberately talkative. What you see is the complete work: a snapshot of the root and of the boot pool, then one clone per dataset — including the `var` children that belong to the boot environment — and finally the result object. `Active` and `Running` are `False` : the new environment exists, is mounted nowhere, and does not boot.

It now appears in the list:

```
Get-OS7BootEnvironment
```

```
Name        : os7_1.0.0.175_202609030759
Release     : 1.0.0.175
Created     : 09/03/2026 09:59:29
Active      : True
Running     : True
Menu        : True
Complete    : True
RootDataset : rpool/ROOT/os7_1.0.0.175_202609030759
BootDataset : bpool/BOOT/os7_1.0.0.175_202609030759
Used        : 4841168896
Origin      :

PS /home/os7admin>
```

> Two boot environments. The new one has no menu entry yet.

`-From` clones another environment instead of the running one; `-Release` gives it a version label.

5.4 Switching

```
Set-OS7BootEnvironment -Name demo
```

That creates the menu entry, sets the EFI pointer and writes `saved_entry` . After the next restart the machine runs out of that environment.

There is no one-shot "just to try it" boot. An environment that is booted without being activated gets the **activated** environment's `/boot` and package database — a half-switched pair that no tool detects. Activation is the only supported switch, and it is reversible; that is what `Restore-OS7` is for.

5.5 Rolling back

The way back after an update that did not work out:

```
Restore-OS7
```

With no parameters this goes back to the previous environment — the one the running environment came from. `-BootEnvironment` names a specific one:

```
Restore-OS7 -BootEnvironment os7_1.0.0.159_202608300312
```

`Restore-OS7` switches, and nothing more. The user's data, the logs, the services' state and everything under `rpool/DATA` and `rpool/USERDATA` stay where they are — they live outside the boot environment and do not roll with it. What is taken back is the system: `/`, `/usr`, `/etc`, the package database and the kernel.

After the restart, a check is worth the seconds:

```
Get-OS7BootEnvironment | Format-Table Name, Active, Running, Menu
Get-OS7Version
```

5.6 Cleaning up

Boot environments take space over time, because each holds the differences from its origin. Remove one that is no longer needed:

```
Remove-OS7BootEnvironment -Name demo -Confirm:$false
```

```
OperationStopped: no boot environment called 'demo'

PS /home/os7admin>
```

Removal clears both halves — the root and the boot pool.

The command asks first; in a script you suppress that with `-Confirm:$false`. Both halves go — an environment with only its boot-pool half left would be a menu entry that leads nowhere.

The running and the active environment cannot be removed.

`Update-OS7` cleans up on its own: `-Keep` decides how many old environments survive a successful update.

5.7 When a machine will not boot

The boot menu holds one entry per complete boot environment. If the machine will not start after an update, pick the previous environment's entry in the menu — the machine comes up, and then you make the switch permanent with `Restore-OS7`.

That is why OS/7 writes its own menu and why one entry per environment is in it: the way back has to work when nothing is running on the machine that you could type a command into.

6 Updates

An update on OS/7 is not a swap of packages in the running system. It is the filling of a **new boot environment** beside the running one, followed by a switch. While the update runs the machine is unchanged and usable; if something goes wrong there is nothing to undo, because nothing was changed.

6.1 What a release is

OS/7 is not delivered as a stream of individual packages but as **curated releases**. Each release is a complete bill of materials: a version, the Ubuntu archive snapshot it was built against, and a version and a hash for every component. Two machines on the same release hold the same packages, regardless of when they were installed.

Releases come from a **signed repository**. The signature is verified not only on download but on the index and on each release descriptor, before anything is listed at all.

6.2 Setting the channel

A freshly installed machine has **no** update source configured. That is deliberate: which source a machine uses, and how quickly it takes releases, is an operational decision.

```
Set-OS7UpdateChannel -Uri https://updates.example.com/os7 -Channel stable
```

Three channels:

Channel	For
`stable`	normal operation
`preview`	validating a release before it is approved
`development`	development builds

Turning the source off without forgetting it:

```
Set-OS7UpdateChannel -Disable
```

6.3 Seeing what is on offer

```
Get-OS7Release
```

```
OperationStopped: /usr/local/share/powershell/Modules/OS7/OS7.Update.ps1:417
Line |
 417 |          throw [System.InvalidOperationException]::new(
     |          ~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~~
     | no release index for channel 'preview' at file:///usr/lib/os7/repo.
     | Either the channel is wrong or this machine is pointed at something that
     | is not an OS/7 repository - see Set-OS7UpdateChannel.

PS /home/os7admin>
```

> Get-OS7Release lists what this machine's channel offers.

`-Available` shows only releases newer than the installed one. `-Channel` and `-Source` query a different channel or source without changing the machine's setting:

```
Get-OS7Release -Available
Get-OS7Release -Channel preview -Source https://updates.example.com/os7
```

Before anything is listed the command verifies the index's signature and each descriptor's hash. A release whose descriptor does not match its hash does not appear — it is not listed with a warning.

6.4 Updating

```
Update-OS7
```

With no parameters this takes the next release on the configured channel. What then happens, in the order it happens:

Update-OS7 — the order that is actually performed

1. Clone the rpool/ROOT and bpool/BOOT pair
2. Assemble the clone and mount it
3. Point apt inside the clone at both repositories
4. apt install os7-<mode>=<version> -> full-upgrade -> autoremove
5. Run the release's migrations (the chroot phase)
6. Rebuild the initramfs, regenerate the menu
7. Activate the new boot environment
8. After the restart: firstboot migrations, prune old environments

The running environment is never

If a step fails, the running environment is still the active one. There is nothing to undo.

> The steps of an update. The running environment is never modified.

The parameters that matter:

Parameter	Effect
`-Version`	a specific release instead of the next one
`-Channel`, `-Source`	once, from a different channel or source
`-Stage`	fill the new environment but do **not** activate it
`-Reboot`	restart immediately after a successful update
`-Keep`	how many old boot environments survive
`-AllowDevelopment`	accept development builds too

A typical maintenance-window sequence:

```
Update-OS7 -Stage                 # during the day: fill, switch nothing
# … in the window:
Set-OS7BootEnvironment -Name os7_1.0.1.0_202609010200
Restart-Computer
```

Or in one go:

```
Update-OS7 -Reboot -Keep 2
```

6.5 After the restart

On the first boot of the new environment the release's **first-boot migrations** run. Those are the steps that can only be done on a running machine — re-sealing the TPM key against the real boot chain, for instance.

Then check the result:

```
Get-OS7Version
Get-OS7BootEnvironment | Format-Table Name, Active, Running, Menu, Release
Get-OS7Service -Detailed | ? { $_.Healthy -eq $false }
```

If something is wrong, one command goes back:

```
Restore-OS7
```

6.6 Checking the decisions without updating

`Test-OS7Update` checks the update path's logic on this machine — with no repository, no ZFS changes and no restart:

```
  PASS  the kernel's mount list is read
  PASS  a nested mount is read too
  PASS  an octal-escaped space is decoded
  PASS  a path that is not mounted is absent
  PASS  an expired index is refused
  PASS  a current index is accepted
  PASS  an index mislabelled as another channel is refused
  PASS  a descriptor matching the index hash is accepted
  PASS  a descriptor whose hash the index does not record is refused
  PASS  migrations are ordered by their numeric prefix
  PASS  a release with no migrations yields none, not an error
  PASS  the firstboot context is separate from the chroot one
  PASS  the dataset serving / is the one at /, not the one mounted elsewhere
  PASS  a machine that is not ZFS-rooted answers null rather than guessing
  PASS  Running decides when the kernel could answer
  PASS  a merely-mounted environment is not the running one
  PASS  and Active is the fallback when it could not
  PASS  a root with nothing mounted under it is refused

OS/7 Update self-test: 31 passed, 0 failed
OS/7 Update self-test: PASS
True

PS /home/os7admin>
```

Test-OS7Update checks the decisions of the update path on this machine.

This is the command for "is this machine fundamentally in order" before you start an update.

6.7 Checking unattended

OS/7 ships a timer that looks periodically for something new on the channel.

The timer is `os7-update-check.timer`, and it is a *scheduled task* — the same distinction Windows makes between a service and a task in the Task Scheduler. Chapter 9.5 has the whole surface; the one command that matters here:

```
Get-OS7ScheduledTask os7-update-check.timer
```

```
OS7-STEP systemd layer: /usr/local/share/powershell/Modules/Systemd/Systemd.psd1

Name        : os7-update-check.timer
Description : Daily OS/7 update check
NextRun     : 09/09/2026 20:01:30
LastRun     : 09/03/2026 10:05:57
LastResult  : success
Activates   : os7-update-check.service
StartupType : disabled
ActiveState : active
SubState    : waiting
Schedule    : *-*-* 00:00:00
Persistent  : True
IsOS7       : True
Healthy     : True


PS /home/os7admin>
```

> The unattended update check, as a scheduled task.

`NextRun` is the answer to "will this fleet stay current without anyone typing anything": a date means yes, an empty field means the timer is not armed — and `Healthy` says so.

The check has a fixed contract on its exit code, and that is what a monitoring system evaluates:

Exit code	Meaning
`0`	nothing to do — or no channel configured
`2`	an update has been staged and is waiting to be switched to
`1`	the check failed

Turning it on and off:

```
Enable-OS7ScheduledTask  os7-update-check.timer
Disable-OS7ScheduledTask os7-update-check.timer
```

6.8 What an update takes with it, and what it does not

The layout from chapter 3 decides, and it is worth repeating because it explains the whole procedure:

Rolls with it: the system itself — `/`, `/usr`, `/etc`, the package database under `/var/lib/dpkg` and `/var/lib/apt`, the package cache, the kernel and the initramfs.

Does not roll with it: the home directories, `/var/log`, `/var/spool`, `/var/tmp`, `/srv`, the services' data under `/var/lib/<service>`, and the management agents' state.

The last of those is the most important and the least obvious. Entra, Intune and Arc hold the other end of a device identity — enrolment records, certificates that rotate on their own schedule, compliance state. The tenant does not roll back. A machine that came back from a rollback presenting a stale identity would be a tenant problem rather than a device problem — and therefore invisible from the device. So that state lives outside the boot environment.

7 Network

Underneath sits **netplan**: a description file says what should hold, and a renderer — `systemd-networkd` on a server, `NetworkManager` on a desktop — carries it out. The OS/7 cmdlets write that description, apply it, and **then ask the kernel** whether anything came of it.

7.1 What is there

```
Get-OS7NetworkAdapter
```

```
OS7-STEP network layer: /usr/local/share/powershell/Modules/Net/Net.psd1

Name         : enp0s2
Kind         : Ethernet
MacAddress   : 52:54:00:12:34:56
OperState    : UP
Up           : True
Carrier      : True
Mtu          : 1500
IPv4Address  : {10.0.2.15/24}
IPv6Address  : {fec0::5054:ff:fe12:3456/64, fe80::5054:ff:fe12:3456/64}
Gateway      : 10.0.2.2
RadioBlocked :


PS /home/os7admin>
```

> The machine's adapters, with what is actually on them.

The output names the adapter, its kind, the MAC address, the link state and the addresses the kernel currently holds. `-Name` asks about one adapter; `-IncludeLoopback` includes `lo`.

7.2 Configured and effective

```
Get-OS7NetworkConfiguration | Format-List
```

```
Renderer            : networkd
RendererIsDefault : False
Files               : {@{Path=/etc/netplan/01-os7-network.yaml; Mode=-rw-------;
                      Bytes=331}, @{Path=/lib/netplan/00-network-manager-all.yaml
                      ; Mode=-rw-------; Bytes=49}}
Configured          : {@{Id=os7net; Kind=Ethernet; MatchMac=; MatchName=en*; Dhcp
                      4=True; Dhcp6=True; Method=Dhcp; Addresses=System.Object[];
                       Gateway=; Nameservers=System.Object[]; SearchDomains=Syste
                      m.Object[]; Ssids=System.Object[]}}
Effective           : {@{Name=enp0s2; Kind=Ethernet; MacAddress=52:54:00:12:34:56
                      ; OperState=UP; Up=True; Carrier=True; Mtu=1500; IPv4Addres
                      s=System.Object[]; IPv6Address=System.Object[]; Gateway=10.
                      0.2.2; RadioBlocked=}}
Disagreements       : {}
Agrees              : True


PS /home/os7admin>
```

> Two separate answers: what netplan says, and what is actually the case on the machine.

This is where the pattern from chapter 1.5 earns the most. The command returns **both, separately**:

- the **configured** side: the merged netplan files, with the files each statement came from named, and which renderer is responsible;
- the **effective** side: the addresses, routes and name servers that actually hold right now.

The machine where the two disagree is the interesting one. A common case: a netplan file describes an adapter this machine does not have — because a disk image came from another device, say. netplan accepts that in silence and configures nothing. OS/7 reports it.

7.3 Configuring an adapter

To DHCP:

```
Set-OS7NetworkAdapter -Name enp0s31f6 -Dhcp
```

To a static address:

```
Set-OS7NetworkAdapter -Name enp0s31f6 `
    -Address 192.0.2.25/24 `
    -Gateway 192.0.2.1 `
    -Nameserver 192.0.2.10, 192.0.2.11 `
    -SearchDomain contoso.local
```

What the command does, and the order is the point:

1. The existing configuration is kept.
2. The new description is written and applied.
3. The command **waits** until the kernel reports an address on the adapter (`-TimeoutSeconds` says how long), and on a static configuration for the gateway as well.

4. If nothing comes up, **the old configuration is put back and applied again**.

That defuses the classic remote-administration accident — a wrong static address does not cut the connection permanently, because the machine goes back on its own.

One thing must never happen quietly: if the rollback also fails, the command says so explicitly. A result carrying `RollbackFailed` is a machine that needs hands at the console — not one to be ignored.

`-Force` skips the confirmation for an adapter the current session is running over.

7.4 Testing reachability

```
Test-OS7Network | Format-List Ok,HasLink,DnsWorks
```

```
Ok       : False
HasLink  : True
DnsWorks : False

PS /home/os7admin>
```

Test-OS7Network checks not only whether the machine is "online" but whether it reaches the services OS/7 exists to reach.

The per-endpoint results hang off the result as a property:

```
(Test-OS7Network).Endpoints | Format-Table -AutoSize
```

```
Name   Host                      Port Reachable Detail
----   ----                      ---- --------- ------
Entra  login.microsoftonline.com  443      True connected in 45 ms
Graph  graph.microsoft.com        443      True connected in 25 ms
Intune manage.microsoft.com       443     False One or more errors occurred. (…
Arc    management.azure.com       443      True connected in 25 ms

PS /home/os7admin>
```

Each endpoint that was tested, with its result.

The command walks the chain — link, gateway, name resolution, endpoints — and says where it breaks. That is the difference between "the network is down" and "DNS answers but the sign-in endpoints are blocked".

Which endpoints exist:

```
Get-OS7Endpoint | Format-Table Name,Host,Port
```

```
Name   Host                      Port
----   ----                      ----
Entra  login.microsoftonline.com  443
Graph  graph.microsoft.com        443
Intune manage.microsoft.com       443
Arc    management.azure.com       443


PS /home/os7admin>
```

The endpoints OS/7 knows about and can test.

The list is a data file beside the module rather than code — sovereign clouds have other host names. `-Cloud` tests against another cloud, `-Endpoint` picks individual ones:

```
Test-OS7Network -Endpoint Entra, Intune
Get-OS7Endpoint -Cloud UsGov
```

7.5 Remote administration with no network

For completeness: if a machine is no longer reachable over the network, the entire command surface works over a serial console. That is not luck but a requirement — every cmdlet in this manual has to be usable on a serial line.

8 Time

On OS/7 the clock is not a side issue but a prerequisite. Kerberos refuses a ticket whose timestamp is more than **five minutes** from the domain controller's clock — and a drifting clock does not report a clock problem. It reports that the password is wrong.

That is why this short chapter comes before the chapters on management and the directory, and why the five-minute limit is built into `Get-OS7TimeSynchronization`.

Time synchronisation is **chrony**, not `systemd-timesyncd`.

8.1 What time is it

```
Get-OS7Time | Format-List
```

```
OS7-STEP time layer: /usr/local/share/powershell/Modules/Time/Time.psd1

LocalTime      : 09/09/2026 19:50:02
UtcTime        : 09/09/2026 17:50:02
TimeZone       : Europe/Berlin
RtcInLocalTime : False
RtcIsDefault   : True
RtcSource      : /etc/adjtime


PS /home/os7admin>
```

The clock: local time, UTC, the zone, and whether the hardware clock is kept in local time.

The last field is the one that matters on a machine that also boots Windows: Windows usually keeps the hardware clock in local time, Linux in UTC. When the two assumptions disagree, the clock jumps at every switch.

8.2 Is the clock being set at all

```
Get-OS7TimeSynchronization | Format-List
```

```
OffsetSeconds        : 5.94E-07
LeapStatus           : Normal
WithinKerberosSkew : True
MaxSkewSeconds       : 300
Servers              : {1.ntp.ubuntu.com, 2.ntp.ubuntu.com, 3.ntp.ubuntu.com, 4.n
                       tp.ubuntu.com…}
Sources              : {@{Mode=Server; State=Selected; Name=185.125.190.122; Stra
                       tum=2; PollInterval=6; Reachability=37; LastReceived=60; O
                       ffsetSeconds=-3.5993E-05; MeasuredOffsetSeconds=-0.0006314
                       69; EstimatedErrorSeconds=0.017099172}, @{Mode=Server; Sta
                       te=Excluded; Name=185.125.190.123; Stratum=2; PollInterval
                       =6; Reachability=37; LastReceived=60; OffsetSeconds=0.0012
                       95733; MeasuredOffsetSeconds=0.001295733; EstimatedErrorSe
                       conds=0.016592225}, @{Mode=Server; State=Excluded; Name=91
                       .189.91.112; Stratum=2; PollInterval=6; Reachability=37; L
                       astReceived=60; OffsetSeconds=0.006531675; MeasuredOffsetS
                       econds=0.006531675; EstimatedErrorSeconds=0.080753148}, @{
                       Mode=Server; State=Excluded; Name=91.189.91.113; Stratum=2
                       ; PollInterval=6; Reachability=37; LastReceived=61; Offset
                       Seconds=0.009386346; MeasuredOffsetSeconds=0.008792523; Es
                       timatedErrorSeconds=0.084359638}…}


PS /home/os7admin>
```

> Whether the clock is being disciplined, by whom, how far off it is — and whether that is close enough for a sign-in to work.

The command distinguishes **three** outcomes, and the distinction is the reason it exists:

Result	Meaning
`$null`	chronyd could not be asked — the service is not running
`$false`	chronyd was asked and is **not** disciplining the clock
`$true`	the clock is being disciplined

"Could not be asked" and "answers no" are different states with different causes. A command that merged both into "not synchronised" would make a crashed service and a missing network connection indistinguishable.

The offset itself comes with it, and the statement whether it is inside the Kerberos tolerance.

8.3 Setting the time zone

```
Set-OS7TimeZone -Id Europe/London
```

The command writes the symlink that actually decides the zone and reads it back. Valid identifiers are the IANA ones (`Europe/London` , `America/New_York` , `UTC`).

8.4 Setting time servers

```
Set-OS7TimeSynchronization -NtpServer dc01.contoso.local, dc02.contoso.local
```

In an Active Directory environment the domain controllers are the right time sources — they are the clock Kerberos measures against.

A pool instead of individual servers:

```
Set-OS7TimeSynchronization -Pool uk.pool.ntp.org
```

`-Exclusive` replaces the existing sources instead of adding to them.

The servers go into a file of their own under chrony's `sources.d` directory, not into `chrony.conf`. chrony is then asked to reload — the service is **not** restarted, because a restart discards the synchronisation already achieved.

8.5 Correcting the clock now

Normally chrony pulls a wrong clock in slowly (*slewing*), so that no time jumps occur. After a long shutdown that is too slow:

```
Sync-OS7Time
```

That tells chrony to step the clock **now**, waits for it to settle (`-SettleSeconds`) and reports what actually happened.

8.6 The order to ask in, on a sign-in problem

When a sign-in against the domain fails with "wrong password" and the password is right, this is the order:

```
Get-OS7TimeSynchronization      # is the clock being set at all?
Get-OS7Time                     # and is it right?
Test-OS7Directory -Domain contoso.local   # measures the skew against the DC
```

`Test-OS7Directory` measures the offset against the domain controller's own clock — and that is the number that counts, not the offset against some time server.

9 Services, scheduled tasks, logs and remoting

9.1 Services

What is a *service* on a Linux machine is a *unit* to **systemd**. `Get-OS7Service` presents them as objects — and answers one question `systemctl` does not.

```
Get-OS7Service -OS7Only -Detailed
```

```
StartupType  : enabled
Result       : success
RestartCount : 0
MainPid      : 912
ActiveSince  : 09/09/2026 17:47:50
IsOS7        : True
ServiceType  : oneshot
Healthy      : True

Name         : zfs-zed.service
Description  : ZFS Event Daemon (zed)
ActiveState  : active
SubState     : running
StartupType  : enabled
Result       : success
RestartCount : 0
MainPid      : 1254
ActiveSince  : 09/09/2026 17:47:52
IsOS7        : True
ServiceType  : simple
Healthy      : True


PS /home/os7admin>
```

The services that make an OS/7 machine, with the health verdict.

`Healthy` is not `active`

In **both** directions. That is why the field exists:

`Healthy = $false` although `systemctl` says `active`. A service in a restart loop has the sub-state `auto-restart`, and `systemctl is-active` reports that as `active`. It is not well.

`Healthy = $true` although the service is not running. A `oneshot` unit — `zfs-mount.service`, for instance — is `inactive/dead` after a successful run, and that is its correct end state. A check that reads "not running" as "sick" reports a fault on a perfectly well machine. The first version of this field did exactly that.

`Healthy` is therefore `$false` when one of these holds:

- the unit is `failed`;
- it is in a restart loop;

- it stopped for a reason that was not success;
- it is enabled at boot, is a service that **stays up**, and is not up.

> **`-Detailed` is required if you want to evaluate `Healthy`.** Without the switch the details were never fetched and `Healthy` is `$null` — not `$false`. A `Where-Object { -not $_.Healthy }` without `-Detailed` therefore matches **every** service.
>
> The correct form is: `powershell Get-OS7Service -Detailed | Where-Object { $_.Healthy -eq $false }`

A single service:

```
Get-OS7Service -Name ssh.service -Detailed | Format-List
```

```
Name         : ssh.service
Description  : OpenBSD Secure Shell server
ActiveState  : inactive
SubState     : dead
StartupType  : disabled
Result       : success
RestartCount : 0
MainPid      : 0
ActiveSince  :
IsOS7        : True
ServiceType  : notify
Healthy      : True


PS /home/os7admin>
```

One service in detail.

Controlling services

```
Start-OS7Service   -Name chrony.service
Stop-OS7Service    -Name chrony.service
Restart-OS7Service -Name chrony.service
```

Each of these reports **what the unit became** — not what was asked of it. A `Start-` that returns an object whose `ActiveState` is not `active` is a failed start, and it looks like one.

Whether a service comes up at boot:

```
Set-OS7Service -Name ssh.service -StartupType Automatic
```

The choices are `Automatic`, `Manual`, `Disabled` and `Blocked`. `Blocked` is stronger than `Disabled`: it also prevents another service from pulling the unit in as a dependency.

9.2 Logs

```
Get-OS7Log -Tail 6 | Format-Table Timestamp,Unit,Message
```

Because these are objects, you filter with PowerShell rather than with `grep`:

```
Get-OS7Log -Priority Error -Since (Get-Date).AddHours(-2)
Get-OS7Log -Unit ssh.service -Tail 50
Get-OS7Log -OS7Only -Boot 0
Get-OS7Log -Since '2026-08-28 14:00' -Until '2026-08-28 15:00' |
    Group-Object Unit | Sort-Object Count -Descending
```

`-Boot 0` is the running boot, `-Boot -1` the one before — useful after an unexpected restart.

Filtering is on the `Unit` field systemd sets itself, not on what the sender claims. Otherwise a program could steer whether it appears in a filtered log.

9.3 The installation log

The installation itself has no journal — the system that would have recorded it did not exist yet. Setup therefore writes a file of its own onto the machine, holding each step's self-proof:

```
Get-OS7InstallLog | Select -Skip 4 -First 8 Message
```

```
Message
-------
2026-09-03 07:59:12 INFO  release: OS/7 1.0.0.175 (preview) (manifest /usr/lib…
2026-09-03 07:59:12 INFO  [not kept]
2026-09-03 07:59:12 INFO  [not kept]
2026-09-03 07:59:12 INFO  unattended: {   "version": 1,   "intent": "Install",…
2026-09-03 07:59:12 INFO  machine: MemTotal 7.7 GiB, MemAvailable 6.6 GiB, Shm…
2026-09-03 07:59:12 INFO  step: Preparing the installer environment
2026-09-03 07:59:12 INFO  installer environment: MemTotal 7.7 GiB, MemAvailabl…
2026-09-03 07:59:12 INFO  ZFS ARC ceiling 0 (the default: half of physical mem…


PS /home/os7admin>
```

The record of the installation, as it sits on the machine.

The file lives at `/var/log/os7-setup/install.log` with mode `0600` and sits **inside** the boot environment — so it survives a rollback, because it belongs to exactly this installation.

Secrets are not in it. Where one was involved the line reads `[not kept]` — it witnesses that something was passed without saying what.

9.4 PowerShell remoting

```
Get-OS7Remoting | Format-List
```

```
OS7-STEP run: /usr/sbin/sshd -T

Subsystem         : True
SubsystemLine     : subsystem powershell /usr/bin/pwsh -sshs -NoLogo
SubsystemReason   :
DropInPath        : /etc/ssh/sshd_config.d/60-os7-powershell.conf
DropInPresent     : True
InteractiveShell  : True
InteractivePath   : /etc/profile.d/95-os7-powershell.sh
Detail            : ssh lands in PowerShell, and Enter-PSSession works


PS /home/os7admin>
```

Whether this machine can be reached with PowerShell — in both senses of the question.

"Reachable with PowerShell" can mean two things, and the command answers both separately:

- **An interactive SSH login lands in PowerShell.** That is true on every OS/7 machine and needs nothing.
- **`Enter-PSSession -HostName` works.** That needs a subsystem in the SSH service which is not configured out of the box.

Turning the second on:

```
Enable-OS7Remoting
```

Then, from another machine:

```
Enter-PSSession -HostName os7-srv-01 -UserName os7admin
Invoke-Command -HostName os7-srv-01 -UserName os7admin -ScriptBlock {
    Get-OS7BootEnvironment
}
```

And off again:

```
Disable-OS7Remoting
```

> **There is no WinRM.** `New-PSSession -ComputerName` exists as a parameter and answers "*no supported WSMan client library was found*". The way onto an OS/7 machine is SSH-based remoting, that is `-HostName`. It is PowerShell's own mechanism and works from Windows too.

9.5 Scheduled tasks

What the Task Scheduler is on Windows, **systemd timers** are here — and they are deliberately not services. `Get-OS7Service` answers "is this program running"; `Get-OS7ScheduledTask` answers "what will run, and when":

```
Get-OS7ScheduledTask | Format-Table Name,NextRun,LastRun
```

```
e2scrub_all.timer               09/13/2026 03:10:33 09/09/2026 19:47:59
fstrim.timer                    09/09/2026 21:17:55 09/03/2026 10:05:57
fwupd-refresh.timer             09/09/2026 20:36:42 09/03/2026 10:05:57
logrotate.timer                 09/09/2026 19:53:41 09/03/2026 10:05:57
man-db.timer                    09/09/2026 22:42:34 09/03/2026 10:05:57
motd-news.timer                 09/09/2026 23:25:21 09/03/2026 10:05:57
os7-backup-replicate.timer      09/09/2026 20:05:12 09/09/2026 19:48:24
os7-update-check.timer          09/09/2026 20:01:30 09/03/2026 10:05:57
sanoid.timer                    09/09/2026 20:00:00 09/09/2026 19:47:52
snapd.snap-repair.timer
sysstat-collect.timer           09/09/2026 19:50:00
sysstat-rotate.timer            09/10/2026 00:00:00
sysstat-summary.timer           09/10/2026 00:07:00
systemd-tmpfiles-clean.timer    09/09/2026 20:02:42
ua-timer.timer
update-notifier-download.timer 09/09/2026 19:52:49
update-notifier-motd.timer      09/14/2026 06:24:32 09/03/2026 10:06:05
zfs-scrub-monthly@.timer
zfs-scrub-weekly@.timer
zfs-trim-monthly@.timer
zfs-trim-weekly@.timer


PS /home/os7admin>
```

Everything that runs on a schedule, with the next and the last run.

The list includes tasks that are currently disabled — a task you have switched off has not ceased to exist, and an inventory that loses it would be an inventory you cannot trust.

Three of these schedules are the product's own: `sanoid.timer` takes the backup snapshots (chapter 12), `os7-backup-replicate.timer` copies them to the backup target, and `os7-update-check.timer` is the unattended update check (chapter 6.7). They are managed like any other task — but they belong to the product, which matters for `Unregister-` below.

One task in full:

```
Get-OS7ScheduledTask sanoid.timer | Format-List
```

```
Name        : sanoid.timer
Description : Run Sanoid Every 15 Minutes
NextRun     : 09/09/2026 20:00:00
LastRun     : 09/09/2026 19:47:52
LastResult  : success
Activates   : sanoid.service
StartupType : enabled
ActiveState : active
SubState    : waiting
Schedule    : *-*-* *:00/15:00
Persistent  : True
IsOS7       : True
Healthy     : True


PS /home/os7admin>
```

> One scheduled task in detail — schedule, next run, last run, and the outcome.

`LastRun` is when the **schedule** last fired; `LastResult` is how that run ended. Starting a task by hand moves `LastResult` and not `LastRun` — the schedule did not fire, and the two answers are kept apart on purpose.

`Healthy`, and the trap it exists to name

A timer can be **enabled and still never run**: systemd's `enable` arms the *next boot* and nothing else. A timer in that state is enabled, inactive, and will not fire until the machine restarts — and no tool on the machine calls that a problem. `Get-OS7ScheduledTask` does: `Healthy` is `$false` and `NextRun` is empty.

`Enable-OS7ScheduledTask` therefore does both — it enables the task *and* arms it now:

```
Enable-OS7ScheduledTask  os7-update-check.timer
Disable-OS7ScheduledTask os7-update-check.timer
```

`Healthy` is also `$false` for a task whose last run failed. As everywhere else, it is `$null` when the details were not fetched — a check that did not run must never read as one that passed.

Running a task now

```
Start-OS7ScheduledTask os7-update-check.timer
```

This starts the task's **service**, waits for it, and reports what happened — the same run the schedule would have produced, just now.

Creating your own task

A weekly pool scrub, Sunday at three in the morning — the parameters no longer fit on one line, so they are splatted, which is the idiom for any parameter set that has outgrown a line:

```
$t = @{ Name='scrub'; Weekly=$true; DayOfWeek='Sunday' }
```

```
PS /home/os7admin>
```

The task's parameters, collected in a hashtable.

```
$t += @{ At='03:00'; Command='Start-ZpoolScrub rpool' }
```

```
PS /home/os7admin>
```

The schedule and the command join them.

```
Register-OS7ScheduledTask @t
```

```
OS7-STEP task written: os7-task-scrub.timer + os7-task-scrub.service

Name        : os7-task-scrub.timer
Description : OS/7 task: scrub
NextRun     : 09/13/2026 03:00:00
LastRun     :
LastResult  : success
Activates   : os7-task-scrub.service
StartupType : enabled
ActiveState : active
SubState    : waiting
Schedule    : Sun *-*-* 03:00:00
Persistent  : False
IsOS7       : True
Healthy     : True


PS /home/os7admin>
```

The registered task, armed — NextRun has a value.

The task is named `os7-task-scrub` — every task you register carries the `os7-task-` prefix, which is what makes it recognisably yours in every listing. `-Command` runs PowerShell; for anything else there is `-Execute` / `-Arguments` with an absolute path. `-Daily -At` and `-Weekly -DayOfWeek -At` cover the common schedules; `-OnCalendar` takes a raw systemd calendar expression for everything they cannot say (`'Mon..Fri 06:30'` , `'*-*-01 06:00:00'`). The schedule is validated **before** anything is written — a spec systemd cannot parse is refused with systemd's own error, not written to disk as a timer that never fires.

Two switches are worth knowing: `-Persistent` catches up after downtime (a machine that was off at three runs the job when it returns), and `-RandomizedDelay` spreads a fleet's runs over a window, so a thousand machines do not start the same job in the same second.

> **No secrets in `-Command`.** The command line lands in a world-readable unit file and shows in systemd's own tooling. A task that needs a credential reads it at run time from a root-owned file with mode `0600`.

Removing a task

```
Unregister-OS7ScheduledTask scrub -Confirm:$false
```

```
OS7-STEP task removed: os7-task-scrub.timer + os7-task-scrub.service

PS /home/os7admin>
```

The task is stopped, disarmed and removed.

`Unregister-` removes **only** tasks that were created with `Register-` — it refuses `sanoid.timer` and the other product schedules by name. Those belong to packages, and deleting a package's unit file leaves the package manager believing in a file that is gone. A product schedule you want silenced is disabled, not removed:

```
Disable-OS7ScheduledTask sanoid.timer
```

9.6 What was not rebuilt

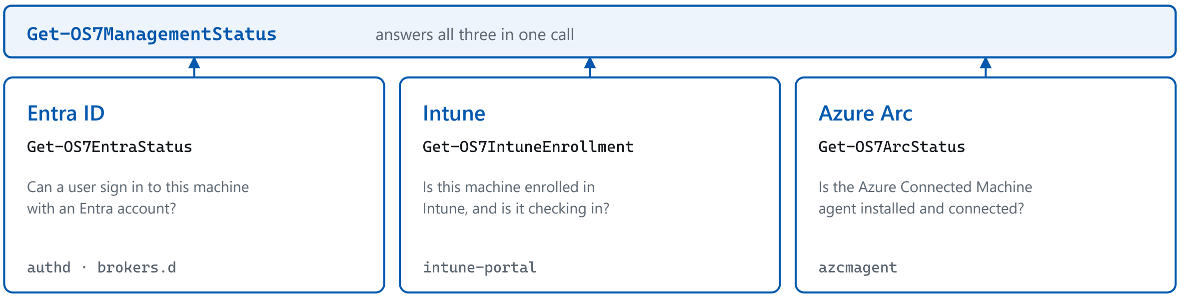
OS/7 rebuilds nothing PowerShell on Linux already does. There is no `Get-OS7Process`, no `Get-OS7FileHash` and no `Test-OS7Connection` — `Get-Process`, `Get-FileHash` and `Test-Connection` work here normally.

The test for admission is not "would this be convenient" but "would an administrator otherwise have to type a Linux command".

`Restart-Computer` was in this paragraph until 2026-09-09, as an example of something that already works. It does not: PowerShell's own version on Linux runs `/usr/sbin/shutdown` with no arguments at all, and that command's default action without a flag is **power off**. So it powered the machine off and reported success. OS/7 therefore supplies `Restart-Computer` and `Stop-Computer` itself. The test for admission has not changed; the answer to "can PowerShell already do this" was wrong.

10 The management plane: Entra ID, Intune and Azure Arc

OS/7 is managed through Microsoft's own tooling. Three of those, and they are **three separate things** with three separate answers:



Three separate paths, three separate answers.

Each of these cmdlets also says WHY something does not work rather than just "no" — a missing broker otherwise looks exactly like a wrong password.

The three management paths and the one command that asks all three.

10.1 All three at once

```
Get-OS7ManagementStatus | Format-List
```

```
             its options are --interactive, --socket-path, --help, --version),
            and no enrolled machine has ever been available to learn the on-dis
            k state from. Enrolment is visible in the Intune portal, not here.;
             Detail=installed and wired; intune-daemon.service is inactive}
Arc        : @{Installed=False; Version=; StagedPackage=System.Object[]; StagedI
            n=/var/cache/os7/packages; Raw=; Reason=; Connected=; Detail=azcmag
            ent is not installed; azcmagent_1.67.03504.1320_amd64.deb is staged
             in /var/cache/os7/packages for the installer to place, which is de
            liberate - its postinst needs a running systemd}
Reachable : False
Endpoints : {@{Name=Entra; Host=login.microsoftonline.com; Port=443; Reachable=
            True; Detail=connected in 59 ms; Verified=}, @{Name=Graph; Host=gra
            ph.microsoft.com; Port=443; Reachable=True; Detail=connected in 32
            ms; Verified=}, @{Name=Intune; Host=manage.microsoft.com; Port=443;
             Reachable=False; Detail=One or more errors occurred. (Name or serv
            ice not known); Verified=}, @{Name=Arc; Host=management.azure.com;
            Port=443; Reachable=True; Detail=connected in 41 ms; Verified=}}
Summary    : authd is installed and PAM is wired to it, but /etc/authd/brokers.d
             is empty - there is no Entra broker for it to talk to, so a sign-i
            n will fail as though the password were wrong. See C8a: authd-msent
            raid is a snap and is not seeded into the image.


PS /home/os7admin>
```

The three paths in one answer, with a summary that names the first cause.

This is the command for the overview. `Summary` does not merely say that something is missing but **why** — which in this corner is the more important half of the answer.

`-SkipNetwork` omits the endpoint reachability check; the command then answers immediately and purely from what is on the machine.

10.2 Entra ID — can anyone sign in here

```
Get-OS7EntraStatus | Format-List
```

```
OS7-STEP run: dpkg-query -W -f=${Status} ${Version} authd
OS7-STEP run: dpkg-query -W -f=${Status} ${Version} microsoft-identity-broker

AuthdInstalled          : True
AuthdVersion            : 0.6.1ubuntu0.1
IdentityBrokerVersion   : 3.0.2-resolute
BrokerRegistered        : False
Brokers                 : {}
BrokerDirectory         : /etc/authd/brokers.d
PamConfigured           : True
ServiceState            : active
ServiceHealthy          : True
Ready                   : False
Detail                  : authd is installed and PAM is wired to it, but /etc/aut
                          hd/brokers.d is empty - there is no Entra broker for it
                           to talk to, so a sign-in will fail as though the passw
                          ord were wrong. See C8a: authd-msentraid is a snap and
                          is not seeded into the image.


PS /home/os7admin>
```

> Whether a user can sign in with an Entra account — and if not, which link of the chain is missing.

Signing in with an Entra account on Ubuntu goes through **authd**, a broker that connects PAM to OpenID Connect. Three things have to hold:

1. `authd` is installed;
2. PAM is wired to `authd`;
3. there is a **broker** — the piece that actually talks to Entra.

The third is the one that is missing in practice, and its absence looks like something else entirely: a sign-in attempt then fails **as though the password were wrong**. That is exactly why `Get-OS7EntraStatus` is the first thing to ask on a machine nobody can sign in to — it names the missing link instead of confirming a guess.

10.3 Intune — is the device enrolled

```
Get-OS7IntuneEnrollment | Format-List
```

```
OS7-STEP run: dpkg-query -W -f=${Status} ${Version} intune-portal

Installed      : True
Version        : 1.2607.4-resolute
Binaries       : {intune-portal, intune-agent, intune-daemon}
PamConfigured  : True
DaemonState    : inactive
DaemonHealthy  : True
SocketPath     : /run/intune/daemon.socket
SocketPresent  : True
Enrolled       :
EnrolledReason : intune-agent exposes no status interface (measured: its option
                 s are --interactive, --socket-path, --help, --version), and no
                  enrolled machine has ever been available to learn the on-disk
                  state from. Enrolment is visible in the Intune portal, not he
                 re.
Detail         : installed and wired; intune-daemon.service is inactive


PS /home/os7admin>
```

What can be said about Intune on this machine — and what cannot.

The command reads what the Intune portal has left on the machine: whether it is installed, whether an enrolment exists, when the device last checked in.

It also says explicitly what it **cannot** answer. A device's compliance state is evaluated in the tenant, not on the device; a machine claiming to be compliant would not know. Where the answer lives only in the tenant, that is said rather than guessed.

Intune enrolment and the portal exist on **amd64**; Microsoft does not ship them for arm64.

10.4 Azure Arc — is the machine inventoried

```
Get-OS7ArcStatus | Format-List
```

```
OS7-STEP run: dpkg-query -W -f=${Status} ${Version} azcmagent

Installed     : False
Version       :
StagedPackage : {azcmagent_1.67.03504.1320_amd64.deb}
StagedIn      : /var/cache/os7/packages
Raw           :
Reason        :
Connected     :
Detail        : azcmagent is not installed; azcmagent_1.67.03504.1320_amd64.deb
                 is staged in /var/cache/os7/packages for the installer to plac
                e, which is deliberate - its postinst needs a running systemd


PS /home/os7admin>
```

Whether the Azure Connected Machine agent is installed, and what it says about itself.

Arc brings a machine into Azure as a resource without it running in Azure — for inventory, policy and update management. `Get-OS7ArcStatus` asks the `azcmagent` agent itself rather than concluding from the existence of a file that it is connected.

> The agent's state lives **outside** the boot environment. A rollback therefore does not take an Arc onboarding back with it — and that is intended, because Azure does not roll back either.

10.5 When management does not take

The order to ask in, because each stage is a prerequisite for the next:

```
Get-OS7TimeSynchronization      # 1. is the clock right?
Test-OS7Network                 # 2. does the machine get out at all?
Test-OS7Network -Endpoint Entra, Intune, Arc   # 3. does it reach the right endpoints?
Get-OS7ManagementStatus         # 4. and what do the three paths say themselves?
```

The clock is deliberately first. A wrong clock brings down every token-based sign-in, and the error it produces points in a completely different direction.

10.6 What is deliberately absent

Group Policy. There is no GPO engine for Linux. A domain-joined machine can **enforce** logon-right policy — that is consumption, not administration.

Anything over RPC or DCOM. `repadmin`, `dcdiag`, `netdom`, DNS and DHCP server management, certificate enrolment: no cross-platform client exists.

11 Active Directory

There are **two entirely different things** you can do with Active Directory, and they are regularly confused. OS/7 keeps them strictly apart:

Administering	Becoming a member
No domain join. No computer account. No new package.	The machine gets a computer account in the domain.
`Enter-OS7AdminSession`	`Join-OS7Domain`
`Get-OS7ADUser · New-OS7ADUser`	`Get-/Set-OS7DomainLogonPolicy`
`Add-OS7ADGroupMember · Search-OS7AD`	`Test-OS7Domain · Repair-OS7Domain`
An administrator signs in to the directory FROM this machine with their own AD account and works as themselves.	That is what lets domain users log in to this machine. Needs adcli and sssd.
`LDAPS 636`	`adcli · sssd · Kerberos`

The two are independent: administering works without membership, membership without an admin session.

> Administering and becoming a member are independent of each other.

Administering means: an administrator signs in to the directory **from** an OS/7 machine with their own AD account and works as themselves — creates users, maintains groups, resets passwords. For that the machine is **not a domain member**, needs no computer account and no extra package. It is an outbound LDAPS connection and nothing else.

Becoming a member means: the machine itself gets a computer account in the domain, so that domain users can sign in **to it**.

If all you want is to maintain directory objects, you do not need the second.

11.1 Preparation: the domain controller's certificate

The connection is **LDAPS on port 636**, and that is not a default but the only route: Active Directory refuses a simple bind on port 389, and the sign-and-seal a Windows client would use instead does not exist on Linux.

For the machine to accept the domain controller's certificate, the issuing authority has to be known:

```
Add-OS7DirectoryTrust -Path /path/to/contoso-ca.crt -Name contoso-issuing-ca
```

The command installs the certificate machine-wide and then **reads back** that it took.

Whether the prerequisites hold:

```
Test-OS7Directory -Domain contoso.local
```

The command walks the chain — DNS, reachability, TLS, bind — and says which link is missing. It also measures the **clock skew against the domain controller's own clock**; see chapter 8.

11.2 Signing in

```
Enter-OS7AdminSession -Domain contoso.local
```

The command asks for credentials and opens a session **for this shell**. `-Credential` passes them ready-made, `-Server` names a particular domain controller instead of the one found through DNS.

```
Get-OS7AdminSession
```

Note what is reported: **the identity the server hands back** — not the name you signed in with. A bind that raised no exception is not proof of who you are; a silent fall back to an anonymous bind would otherwise look like a success.

Closing the session:

```
Exit-OS7AdminSession
```

> **The credential is used and then forgotten.** There is no session that can re-authenticate itself later — that would be a password at rest. After the shell restarts you sign in again.

11.3 Users

```
Get-OS7ADUser -Identity jsmith
Get-OS7ADUser -Filter '(department=Sales)' -SearchBase 'OU=London,DC=contoso,DC=local'
Get-OS7ADUser -Identity jsmith -Property department, manager, lastLogonTimestamp
```

`Enabled` and `LockedOut` are **separate answers from separate places**. A locked account is not a disabled one, and confusing them leads to remedies that change nothing.

Creating an account:

```
$pw = Read-Host -AsSecureString -Prompt 'Password'
New-OS7ADUser -Name jsmith `
    -Path 'OU=London,DC=contoso,DC=local' `
    -DisplayName 'Jane Smith' `
    -GivenName Jane -Surname Smith `
    -UserPrincipalName jsmith@contoso.local `
    -Mail jsmith@contoso.local `
    -Password $pw -Enabled
```

The order behind that is forced rather than chosen: Active Directory creates an account **disabled**, then takes the password, and only then lets it be enabled. An account asked for with a password and enabled in one go is rejected.

Changing, disabling, unlocking:

```
Set-OS7ADUser -Identity jsmith -Department Sales
Disable-OS7ADAccount -Identity jsmith
Enable-OS7ADAccount  -Identity jsmith
Unlock-OS7ADAccount  -Identity jsmith
```

`Enable-` and `Disable-` read the existing flags and change **one bit**. They do not rewrite the whole value, so nothing else is discarded along the way.

Resetting a password:

```
$pw = Read-Host -AsSecureString -Prompt 'New password'
Reset-OS7ADAccountPassword -Identity jsmith -NewPassword $pw -MustChangeAtNextLogon
```

Over an unencrypted connection the command refuses **before** the password reaches the wire.

11.4 Groups

```
Get-OS7ADGroup -Identity 'Sales-London'
Get-OS7ADGroupMember -Identity 'Sales-London'
Get-OS7ADGroupMember -Identity 'Sales-London' -Recursive

New-OS7ADGroup -Name 'Sales-London' -Path 'OU=Groups,DC=contoso,DC=local' -Scope Global

Add-OS7ADGroupMember    -Identity 'Sales-London' -Member jsmith, rjones
Remove-OS7ADGroupMember -Identity 'Sales-London' -Member rjones
```

`-Recursive` uses the directory's own matching rule rather than rebuilding the nesting in the client. And `Add-OS7ADGroupMember` adds — it does not replace the membership list.

11.5 Computers and organisational units

```
Get-OS7ADComputer -Identity WS-LON-014
Get-OS7ADComputer -Filter '(operatingSystem=*Server*)'
Get-OS7ADOrganizationalUnit -SearchBase 'DC=contoso,DC=local'
Move-OS7ADObject -DistinguishedName 'CN=WS-LON-014,CN=Computers,DC=contoso,DC=local' `
                 -TargetPath 'OU=London,DC=contoso,DC=local'
Rename-OS7ADObject -DistinguishedName 'CN=old,OU=…' -NewName new
```

Nobody has to type a computer account's trailing `$`.

11.6 When the curated surface is not enough

A directory surface you cannot escape from would have to be complete. So the raw access sits beside it:

```
Search-OS7AD -Filter '(&(objectClass=user)(!
(userAccountControl:1.2.840.113556.1.4.803:=2)))' `
             -SearchBase 'DC=contoso,DC=local' -Scope Subtree

Get-OS7ADObject -DistinguishedName 'CN=Jane Smith,OU=London,DC=contoso,DC=local'
Set-OS7ADObject -DistinguishedName 'CN=Jane Smith,OU=…' -Name extensionAttribute1 -Value
'X'
Remove-OS7ADObject -DistinguishedName 'CN=old,OU=…'
```

Searches follow the pages to the end — a result is never silently cut off at 1000 objects.

11.7 Joining a domain

So that **domain users can sign in to this machine**, it joins the domain:

```
$pw = Read-Host -AsSecureString -Prompt 'Password of the join account'
Join-OS7Domain -Domain contoso.local `
    -UserName joiner `
    -Password $pw `
    -OrganizationalUnit 'OU=Linux,OU=Servers,DC=contoso,DC=local' `
    -AllowGroup 'CONTOSO\Linux-Users' `
    -AdministratorGroup 'CONTOSO\Linux-Admins'
```

`-AllowGroup` decides who may sign in; `-AdministratorGroup` who may administer the machine. The `sudo` rule that results is checked by `visudo` before it counts — a malformed rule would otherwise break every elevation on the machine.

The password goes to the join tool on **stdin**, never as an argument; an argument would sit in the process list.

State and verification:

```
Get-OS7Domain | Format-List
Test-OS7Domain -Domain contoso.local
```

```
OS7-STEP directory layer: /usr/local/share/powershell/Modules/Directory/Director
y.psd1

ConfiguredDomains     : {}
ConfigurationPresent : False
KeytabPresent        : False
KeytabPrincipal      : {}
Krb5ConfPresent      : True
SssdRunning          : False
ProbeAccount         :
ProbeResolved        :
Joined               : False
Detail               : {There is no /etc/sssd/sssd.conf, so this machine is not
                        configured for any domain., A domain join does not make
                        this machine manageable by Intune: enrolment goes throu
                       gh Entra ID, and there is no hybrid join for Linux. Grou
                       p Policy is not applied either.}


PS /home/os7admin>
```

Whether this machine is a member of a domain — as configured and as it actually stands.

`Get-OS7Domain` again answers both halves: what is configured, and what actually holds. `Test-OS7Domain` checks whether the membership **works** — whether a directory account resolves through the system's name service. That is the only question whose answer proves a working join.

Changing the logon policy afterwards:

```
Get-OS7DomainLogonPolicy
Set-OS7DomainLogonPolicy -AdministratorGroup 'CONTOSO\Linux-Admins'
```

After a **rollback** the machine account can be stale, because the domain has changed the password since and the boot environment brought the old state back:

```
Repair-OS7Domain -Domain contoso.local
```

Leaving:

```
Remove-OS7Domain -Domain contoso.local -UserName joiner -Password $pw
```

11.8 Kerberos tickets

```
Get-OS7KerberosTicket
New-OS7KerberosTicket -Principal administrator@CONTOSO.LOCAL
Remove-OS7KerberosTicket
```

```
Known Reason              Principal Ticket
----- ------              --------- ------
 True no credential cache           {}


PS /home/os7admin>
```

This session's Kerberos tickets.

The realm is upper-case — that is Kerberos convention, not a choice.

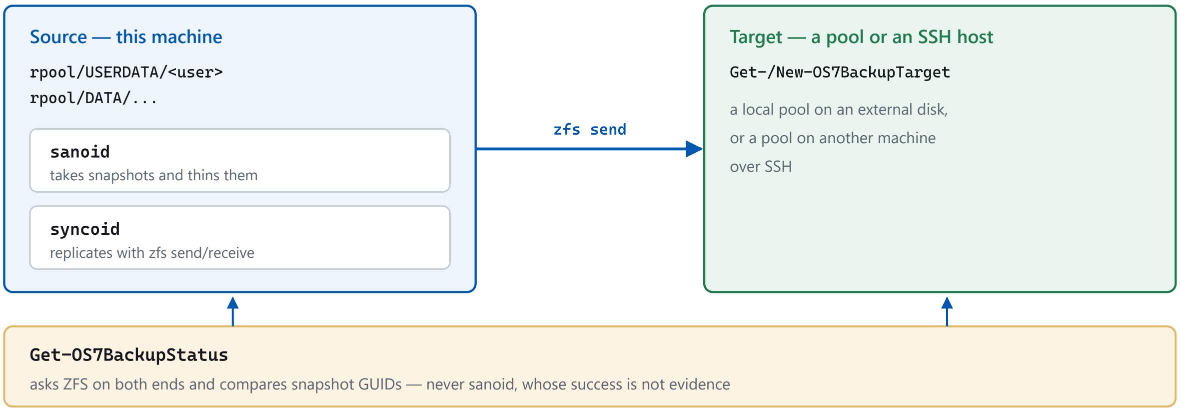
> **Domain users' home directories live outside the boot environment**, exactly like the local ones. Otherwise a rollback would take signed-in domain users' files back with it.

12 Backup and restore

OS/7's backup has two halves, and the split explains the whole chapter:

- **Snapshots on this machine** protect against mistakes — deleted by accident, overwritten wrongly, a migration that went badly. They cost almost nothing and are available immediately.
- **Replication to a target** protects against losing the machine. It needs a second medium and time.

Snapshots are on by default; replication is a deliberate decision.



The backup architecture: sanoid snapshots, syncoid replicates, and OS/7 verifies — by asking ZFS, never the tools.

> **Why OS/7 verifies.** Both tools report success in cases where nothing happened: sanoid exits 0 even when taking a snapshot failed, and its own monitoring answers from a cache that is deliberately allowed to be five hours old. So `Get-OS7BackupStatus` asks **ZFS** — on the source, and over SSH on the target — and neither of the two tools.

12.1 Looking at the policy

```
Get-OS7BackupPolicy | Format-List
```

```
Configured  : True
Enabled     : True
ConfigPath  : /etc/os7/backup.json
SanoidConf  : /etc/sanoid/sanoid.conf
Sources     : {@{Dataset=rpool/USERDATA; Recursive=True; ChildrenOnly=True; Ret
              ention=System.Collections.Specialized.OrderedDictionary; Exists=T
              rue; Datasets=System.Object[]; Snapshots=8; Newest=09/09/2026 19:
              47:53}, @{Dataset=rpool/DATA/srv; Recursive=True; ChildrenOnly=Fa
              lse; Retention=System.Collections.Specialized.OrderedDictionary;
              Exists=True; Datasets=System.Object[]; Snapshots=4; Newest=09/09/
              2026 19:47:53}}
TargetCount : 0


PS /home/os7admin>
```

> What this machine snapshots and how long it keeps it.

The detail per dataset:

```
(Get-OS7BackupPolicy).Sources | Format-Table Dataset, Snapshots, Newest
```

The policy lives in `/etc/os7/backup.json`. That file is the authority; the configuration for sanoid is generated from it — and parsed by sanoid itself before it is installed.

12.2 Changing the policy

Which datasets are backed up:

```
Set-OS7BackupPolicy -Dataset rpool/USERDATA, rpool/DATA/srv
```

How long the snapshots are kept:

```
Set-OS7BackupPolicy -Retention @{ hourly = 48; daily = 30; weekly = 8; monthly = 12 }
```

On and off:

```
Enable-OS7Backup
Disable-OS7Backup
```

Disabling **destroys nothing**. Existing snapshots stay; no new ones are taken and no old ones are pruned.

> **`rpool/ROOT` and `bpool/BOOT` do not belong in the policy.** Those datasets belong to the boot-environment mechanism. The attempt is refused — a second thing taking and pruning snapshots there would get in the rollback's way.

12.3 What the policy actually reaches

A policy that names a home directory which is not on a dataset of its own does not back that home up — it backs nothing up under that name, and does not say so by itself:

```
Get-OS7BackupCoverage | Format-List
```

```
Path       : /home/os7admin
Dataset    : rpool/USERDATA/os7admin_d5712bc8
OwnDataset : True
Covered    : True
Reason     : covered

PS /home/os7admin>
```

> Which home directories the policy actually reaches — and for those it does not, why.

Only what is not covered:

```
Get-OS7BackupCoverage | Where-Object { -not $_.Covered }
```

This is the companion to `Get-OS7Home` in chapter 4.4: there the question was whether a home is on a dataset of its own; here it is whether the backup reaches it. A home without its own dataset cannot be caught by any snapshot policy.

12.4 Setting up a target

An external disk, prepared by OS/7:

```
$pw = Read-Host -AsSecureString -Prompt 'Passphrase for the backup disk'
New-OS7BackupTarget -Name usb `
    -CreateOn /dev/disk/by-id/usb-Samsung_T7_0001 `
    -ConfirmDisk /dev/disk/by-id/usb-Samsung_T7_0001 `
    -Passphrase $pw
```

> **`-ConfirmDisk` is not a prompt, it is a second naming.** This command **erases the named disk**. Someone who has to type it twice does not type the wrong one — a yes/no prompt only confirms that the command was submitted.

A pool on another machine, over SSH:

```
New-OS7BackupTarget -Name nas -ComputerName backup@nas.lan -Dataset tank/os7
```

An existing local pool:

```
New-OS7BackupTarget -Name second -Pool tank -Dataset tank/os7
```

Looking at targets and testing them:

```
Get-OS7BackupTarget | Format-Table Name, Kind, Present, InSync, NewestReplicated
Test-OS7BackupTarget usb
```

Mounting and unmounting a removable disk:

```
Mount-OS7BackupTarget usb -Passphrase $pw
# … replication …
Dismount-OS7BackupTarget usb
```

`Dismount-` exports the pool and closes the LUKS container so the disk can be pulled safely.

12.5 Backing up

Snapshots run on a schedule. By hand:

```
Start-OS7Backup                         # snapshots only
Start-OS7Backup -Replicate              # snapshots, then replicate
Start-OS7BackupReplication -Target nas  # replicate only, this target only
```

The replication run takes a lock, because syncoid has none of its own — two concurrent runs against the same source would be a problem.

12.6 Is it really backed up

```
Get-OS7BackupStatus | Format-List
```

The command asks ZFS on the source and, through the ZFS layer over SSH, on the target — and compares the snapshots' **GUIDs**. Not their names: two snapshots with the same name are not the same object, and a name comparison would treat a broken replication chain as sound.

`-SkipTargets` limits the answer to the local side; that is fast and needs no reachable target.

For monitoring:

```
Get-OS7BackupStatus | ConvertTo-Json -Depth 6
```

12.7 Getting a file back

The most common case of all: a single file from yesterday.

```
Get-OS7FileVersion /home/jsmith/quote.docx
```

The command walks the snapshots and lists every version still held, with its time and size. `-DistinctOnly` hides consecutive identical versions, `-Newest` limits the count, `-IncludeCurrent` brings the live file into the comparison.

Restoring:

```
# yesterday's 18:00 version, back to where it came from
Restore-OS7File /home/jsmith/quote.docx -AsOf '2026-08-28 18:00'

# a specific version, somewhere else - safer, because nothing is overwritten
Restore-OS7File /home/jsmith/quote.docx `
    -Snapshot autosnap_2026-08-28_18:00:01_hourly `
    -Destination /home/jsmith/quote-old.docx
```

Without `-Destination` the existing file is replaced, and for that the command requires `-Force`.

> A user who owns the file needs no administrator for this: snapshots are readable to them, and `Get-OS7FileVersion` runs without elevation.

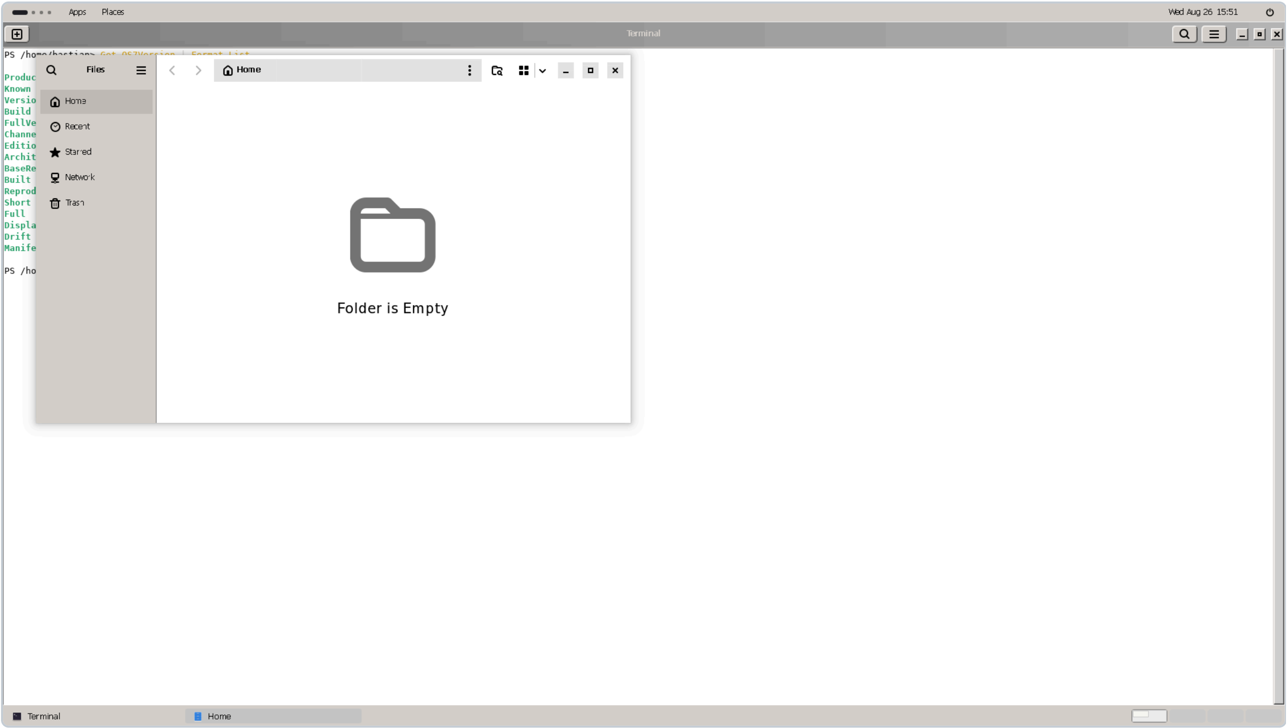
13 The desktop

The desktop belongs to the **amd64** edition of OS/7; there is none on arm64, because Microsoft ships no desktop stack there. It is chosen during installation (screen 8, "GUI").

13.1 What is on it

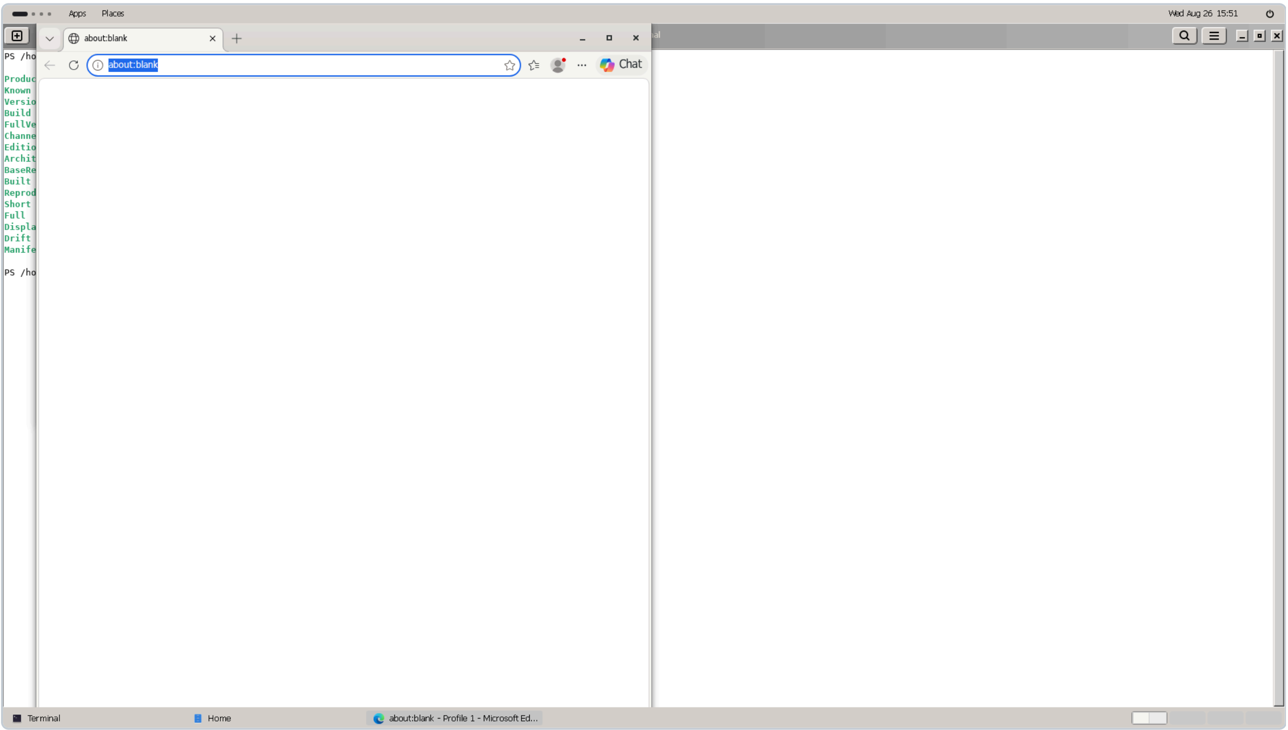
The OS/7 desktop in its classic appearance.

Technically it is GNOME. Appearance and behaviour are brought onto the classic Windows pattern: a bar at the bottom, a start menu on the left, a notification area on the right, square windows with visible frames.

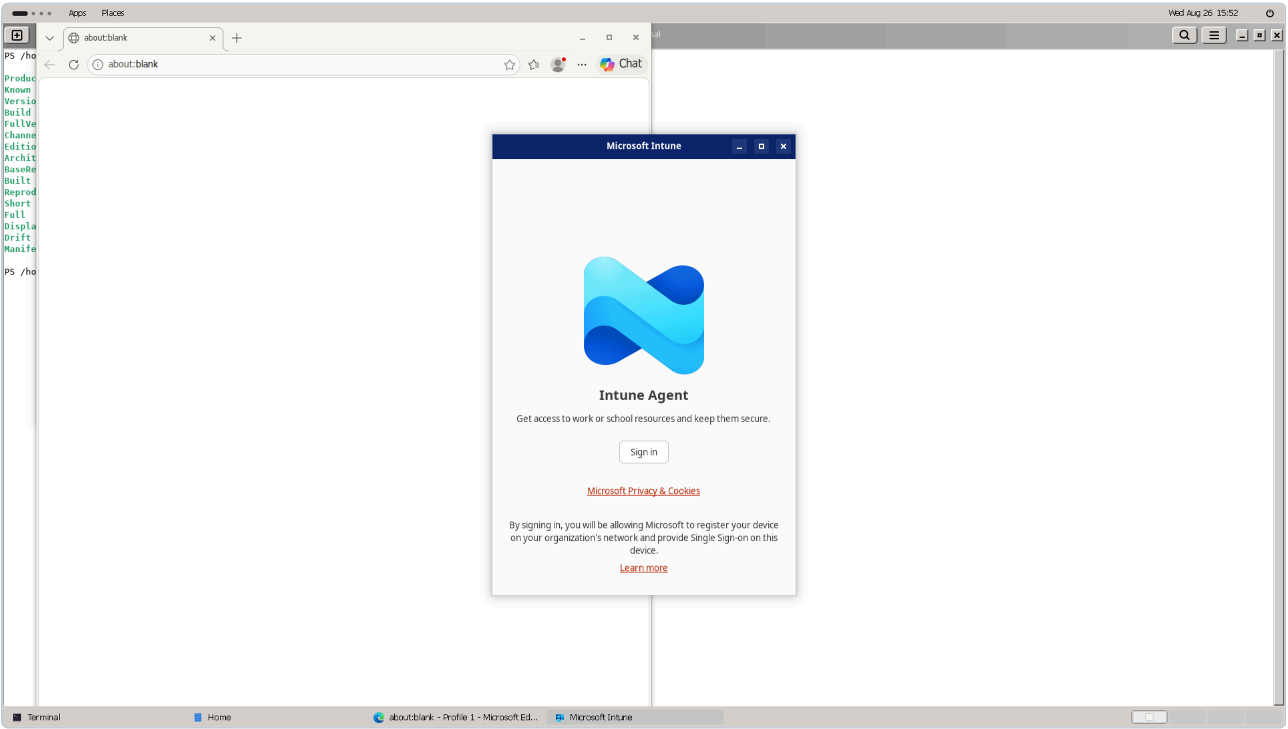


The file manager.

Microsoft Edge ships as the default browser, along with the **Intune Company Portal**:



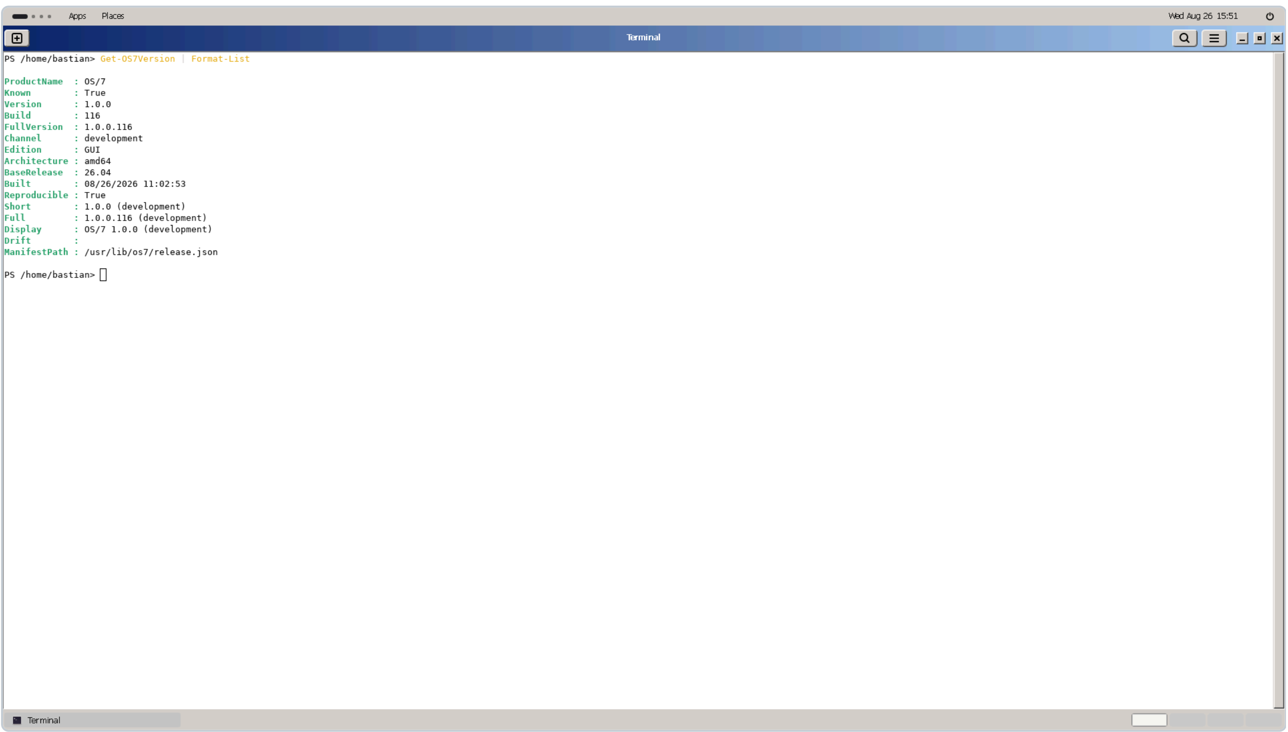
Microsoft Edge.



The Intune Company Portal.

13.2 The terminal window

A terminal window on the desktop lands in **PowerShell**, exactly like the console and an interactive SSH login:



A terminal window on the desktop: PowerShell, as everywhere else.

That is arranged explicitly and is not automatic. A terminal emulator normally starts the shell **not** as a login shell, and the file that holds the hand-off to PowerShell is read only by login shells. On OS/7 the terminal's default profile is therefore set to "run as a login shell" — so the same hand-off runs as everywhere else.

13.3 Switching the appearance

```
Get-OS7Theme
Set-OS7Theme -Name Classic     # the classic OS/7 appearance
Set-OS7Theme -Name Stock       # untouched GNOME
```

The switch applies to the **current session** and takes full effect at the next sign-in.

`Stock` is not merely decorative: when a problem appears on the desktop, switching to `Stock` separates "is it the appearance" from "is it GNOME" — and on a rendering problem that is the first useful bisection.

13.4 The sign-in screen and the console

The sign-in screen is branded too. It reads its settings from a **different** database than the signed-in session — anyone changing something there changes it in two places or in none.

The text console uses a different font from the installer: the installer shows Fixedsys, the installed console **Cascadia Mono**. That is deliberate — Setup should look like an installer from that era, the installed machine like a working tool of today.

13.5 What is not different on the desktop

Everything in this manual. There is no graphical administration surface that would open a second set of rules beside the cmdlets: boot environments, updates, network, backup and the directory are driven on a desktop machine with the same commands as on a server.

14 Diagnostics

This chapter is an order, not a list of commands. The commands are in the chapters before it; here is the sequence to ask them in, and why the sequence is what it is.

14.1 The rule underneath

Ask the thing itself, not its exit code.

The expensive faults all have the same shape: a program reported success, and the thing it was meant to change did not change. An exit code is a diagnostic. So is a log line.

In practice: when a command says it wrote the network configuration, ask the kernel which address is on the adapter. When a service reports `active`, ask for `Healthy`. The cmdlets in this manual do that by themselves — the rule is for everything you do beside them.

And: a diagnostic must not depend on the subsystem it is diagnosing. The state of the network cannot be asked over the network; the state of systemd not through a service systemd starts.

14.2 First look at a machine

```
Get-OS7Version                                  # which OS/7, which boot environment
Get-OS7BootEnvironment | Format-Table Name, Active, Running, Menu, Release
Get-Zpool | Format-Table Name, Health, Free
Get-OS7Service -Detailed | Where-Object { $_.Healthy -eq $false }
Test-OS7Network
Get-OS7ManagementStatus
```

Six commands, about a minute. If nothing here stands out, the problem is not in the machine's foundations.

14.3 A sign-in fails

The most common wrong turn: the password is assumed to be wrong, because that is what the error says.

```
# 1. The clock. Kerberos refuses beyond five minutes of skew,
#    and the message then reads "wrong password".
Get-OS7TimeSynchronization
Get-OS7Time

# 2. With Entra: is the broker missing? That also looks like a wrong password.
Get-OS7EntraStatus

# 3. With a domain: which link of the chain is missing?
Test-OS7Directory -Domain contoso.local
Test-OS7Domain    -Domain contoso.local

# 4. Only now, the log.
Get-OS7Log -Priority Error -Since (Get-Date).AddMinutes(-15)
```

The order is chosen so that the causes which disguise themselves as something else are checked first.

14.4 No network

```
Get-OS7NetworkAdapter                  # is there an adapter, does it have link?
Get-OS7NetworkConfiguration            # configured and effective - do they agree?
Test-OS7Network                        # where does the chain break?
```

The second is the telling one. A common case is a netplan file describing an adapter this machine does not have — netplan accepts that in silence and configures nothing.

After a failed change:

```
Set-OS7NetworkAdapter -Name enp0s31f6 -Dhcp
```

And if an earlier attempt ended in `RollbackFailed`, the machine needs hands at the console — that is the case where the rollback itself failed.

14.5 The machine will not boot

1. In the boot menu, pick the entry for the **previous boot environment**. There is one for every complete environment.
2. Once the machine is up, make the switch permanent: `powershell Restore-OS7`
3. Then look at what was wrong in the new environment: `powershell Get-OS7Log -Boot -1 -Priority Error`

If the machine drops to an initramfs prompt, the disk was not unlocked or the pool was not imported. That is the corner where the passphrase is needed — the TPM unlock does not take when the boot chain has changed, after a firmware update for instance.

14.6 An update went wrong

```
Get-OS7BootEnvironment | Format-Table Name, Active, Running, Menu, Complete, Release
```

Read `Running`, not `Active` — during an update two environments are `Active` and only one is running.

`Complete = False` means an environment is missing one of its halves. Such an environment should not be booted; remove it and start the update again.

Back:

```
Restore-OS7
```

14.7 The pool is full

```
Get-ZfsSpace rpool | Format-List
Get-ZfsDataset | Sort-Object Used -Descending | Select-Object -First 10 Name, Used
Get-ZfsSnapshot | Sort-Object Used -Descending | Select-Object -First 10 Name, Used
Get-OS7BootEnvironment | Format-Table Name, Used, Running
```

Three usual causes, in that frequency:

1. **Old boot environments.** Each holds the differences from its origin. `Remove-OS7BootEnvironment` clears them, and `Update-OS7 -Keep` stops them accumulating.
2. **Backup policy snapshots.** Retention is set with `Set-OS7BackupPolicy -Retention`.
3. **Hand-made snapshots** nobody removed.

14.8 What to attach to a report

When you pass a fault on, attach this — it answers most of the follow-up questions in advance:

```
Get-OS7Version -Detailed
Get-OS7BootEnvironment | Format-Table *
Get-OS7Service -Detailed | Where-Object { $_.Healthy -eq $false } | Format-Table *
Get-OS7Log -Priority Error -Boot 0 | Select-Object -Last 50
Get-OS7InstallLog | Select-Object -Last 40
```

The installation log is particularly useful, because it holds each installation step's self-proof — and because it survives a rollback.

Appendix A Every command

This list was **generated by asking the modules** — every name, every description and every parameter list comes from `Get-Command` and `Get-Help` on an OS/7 machine. What is missing here is missing from the product; what is here exists with exactly these parameters.

Mandatory parameters are bold. A cmdlet with several parameter sets has several mandatory parameters, of which only one is needed at a time — `Get-Help <command>` shows the sets separately.

The six modules

Module	Layer	Commands
`powershell/Zfs/`	generic	26
`powershell/Net/`	generic	11
`powershell/Time/`	generic	9
`powershell/Systemd/`	generic	21
`powershell/Directory/`	generic	36
`powershell/OS7/`	the product	140

`Zfs`

ZFS: pools, datasets, snapshots. Knows nothing about OS/7.

Command	What it does	Parameters
`Clear-ZfsProperty`	Return a property to its inherited or default value. `zfs inherit`.	**Name**, **Property**, Recurse
`Clear-ZpoolLabel`	Remove ZFS's label from a device, so a pool can be created on it again.	**Device**, Force
`Convert-ZfsClone`	Promote a clone so it no longer depends on its origin. `zfs promote`.	**Name**
`Dismount-ZfsDataset`	Unmount a dataset. `zfs unmount`.	Name, All, Force
`Export-Zpool`	Export a pool, so it can be imported elsewhere.	**Name**, Force
`Format-ZfsSize`	A byte count as a human-readable binary size. The display half of Z6.	Size
`Get-ZfsDataset`	Filesystems, volumes and snapshots, as objects.	Name, Type, Recurse, Depth, Property, ComputerName, SshArgument
`Get-ZfsProperty`	Properties of a dataset, one object per property, with their source.	**Name**, Property
`Get-ZfsSnapshot`	Snapshots, as objects.	Name, NoRecurse, ComputerName, SshArgument
`Get-ZfsSpace`	Where a dataset's space has actually gone.	Name, Recurse
`Get-Zpool`	The ZFS storage pools on this system.	Name, ComputerName, SshArgument
`Get-ZpoolStatus`	Pool health with the vdev tree as objects, not as indented text.	Name, Flat
`Import-Zpool`	Import a pool.	Name, All, Directory, AltRoot, Force
`Mount-ZfsDataset`	Mount a dataset, and confirm from ZFS that it is mounted.	Name, All, Overlay
`New-ZfsClone`	Create a writable clone of a snapshot.	**Snapshot**, **Name**, Property, Parents
`New-ZfsDataset`	Create a filesystem or a volume.	**Name**, Property, Size, Parents, Sparse
`New-ZfsSnapshot`	Snapshot a dataset.	**Name**, **SnapshotName**, Recurse, Property
`New-Zpool`	Create a storage pool.	**Name**, **Device**, Layout, Property, FilesystemProperty, AltRoot, Force
`Remove-ZfsDataset`	Destroy a dataset. Irreversible.	**Name**, Recurse
`Remove-ZfsSnapshot`	Destroy a snapshot. Irreversible.	**Name**, Recurse

Command	What it does	Parameters
`Remove-Zpool`	Destroy a pool and everything in it. Irreversible.	**Name**, Force
`Rename-ZfsDataset`	Rename or move a dataset.	**Name**, **NewName**, Parents, Force
`Restore-ZfsSnapshot`	Roll a dataset back to a snapshot. DESTRUCTIVE.	**Name**, Force, DestroyClones
`Set-ZfsProperty`	Set one or more properties on a dataset, and report what ZFS then holds.	**Name**, **Property**, **PropertyName**, **Value**
`Start-ZpoolScrub`	Start, pause or stop a scrub, and report what the pool then says.	**Name**, Stop, Pause
`Test-ZfsModule`	Check the module against recorded real ZFS output, or against real ZFS.	Live, LiveWrite, Pool, FixturePath

`Net`

The network: `ip`, netplan, networkd, NetworkManager, resolved.

Command	What it does	Parameters
`Get-NetLink`	The network links this kernel has, with the addresses actually on them.	Name, SysfsRoot
`Get-NetplanConfiguration`	What netplan is configured to do — merged, with the files it was merged from named separately.	RootDir
`Get-NetRadio`	Whether any radio is blocked — and whether that question could be answered at all.	*none*
`Get-NetRoute`	The routes this kernel has.	Family
`Invoke-NetplanApply`	Runs `netplan apply`. DOES NOT JUDGE WHETHER IT WORKED.	*none*
`New-NetplanDocument`	Renders a netplan document. A pure function — it touches no file, no interface and no running system.	**Renderer**, **Version**, WrittenBy, InterfaceName, MacAddress, Wireless, **Method**, Address, Gateway, Nameserver, SearchDomain, Ssid, Hidden, Security, Psk, Identity, AnonymousIdentity, Password, CaCertificate
`Remove-NetplanDocument`	Deletes a netplan document. Used to undo a document that did not exist before.	**Path**
`Resolve-NetSrvRecord`	The SRV records for a name — and whether the resolver could be asked at all.	**Name**, Server, Tool
`Set-NetplanDocument`	Writes a netplan document, and hands back what was there before.	**Path**, **Content**
`Test-NetModule`	Checks this module's own contract. Needs no network and no root.	FixturePath
`Wait-NetLinkAddress`	Waits for a link to have an address, and answers by asking the kernel.	**Name**, TimeoutSeconds, RequireGateway

`Time`

The clock: chrony, the time zone, the hardware clock.

Command	What it does	Parameters
`Get-ChronySource`	The time sources chrony knows about, and what it thinks of each.	*none*
`Get-ChronySourceFile`	Where chrony reads NTP sources from, and what is in each file.	Root
`Get-ChronyTracking`	What chrony is doing with this clock — whether it is disciplined, by whom, and how far off it is.	*none*
`Get-SystemClock`	The clock: local, UTC, and whether the hardware clock is kept in local time.	Root
`Get-SystemTimeZone`	The zone this machine is in, read from the only thing that decides it.	Root
`Set-ChronySource`	Writes one `.sources` file and asks chrony to reload — without restarting it.	**Name**, Server, Pool, Root
`Set-SystemTimeZone`	Sets the zone by writing the symlink, and reads it back.	**Id**, Root
`Sync-ChronyClock`	Tells chronyd to STEP the clock now rather than slew it.	*none*
`Test-TimeModule`	Checks this module against RECORDED REAL chrony output and a zone tree it builds. No daemon, no root, no network.	FixturePath

`Systemd`

Units and the journal.

Command	What it does	Parameters
`Get-SystemdHostName`	The static, transient and pretty host names, from the places that hold them.	*none*
`Get-SystemdJournal`	The journal, as objects with real types.	Unit, Identifier, Priority, Since, Until, Tail, Boot
`Get-SystemdSession`	Who is signed in to this machine, and how.	Id, User, Remote, Class
`Get-SystemdTimer`	The timers this machine has: what they will run, when they will next run, and when they last did.	Name, Detailed
`Get-SystemdUnit`	The units this machine has, and what state they are in.	Name, Type, State, Detailed
`Get-SystemdUnitFreezerState`	`running`, `freezing`, `frozen` or `thawing` for one unit — or $null when systemd would not say.	**Name**
`Invoke-SystemdShutdown`	Reboots, powers off or halts this machine — the action always spelled out.	**Action**, Delay, Message
`New-SystemdService`	Writes a .service unit and asks systemd whether it loaded.	**Name**, **Command**, Description, DependsOn, User, Enabled, Force
`New-SystemdTimer`	Writes a timer and the service it activates, and asks systemd whether it loaded them.	**Name**, **Command**, **OnCalendar**, Description, User, Persistent, RandomizedDelay, Force
`Remove-SystemdService`	Removes a .service unit file this module could have written, and asks systemd whether it is gone.	**Name**
`Remove-SystemdTimer`	Removes a timer unit pair this module could have written, and asks systemd whether it is gone.	**Name**
`Restart-SystemdUnit`	Restarts a unit and reports what it became.	**Name**
`Resume-SystemdUnit`	Thaws a frozen unit's processes, and reads the freezer back.	**Name**
`Set-SystemdHostName`	Sets the machine's name — static and transient together — and reads it back from the kernel.	**Name**, Pretty, SkipHostsFile
`Set-SystemdUnitStartup`	Whether a unit starts at boot, and reports what it became.	**Name**, **Startup**
`Start-SystemdUnit`	Starts a unit and reports what it became.	**Name**
`Stop-SystemdSession`	End a session: everything running in it is stopped.	**Id**, Force

Command	What it does	Parameters
`Stop-SystemdUnit`	Stops a unit and reports what it became.	**Name**
`Suspend-SystemdUnit`	Freezes a unit's processes, and reads the freezer back.	**Name**
`Test-SystemdModule`	Checks this module against RECORDED REAL systemctl and journalctl output. Needs no systemd, no root and no journal.	FixturePath
`Update-SystemdUnit`	Tells a running unit to re-read its own configuration — `systemctl reload`.	**Name**

`Directory`

LDAP, Kerberos and realm membership. Knows nothing about Active Directory's policy and nothing about OS/7.

Command	What it does	Parameters
`Connect-DirectoryServer`	Open a bound connection to an LDAP server and return a session object.	**Server**, Credential, Port, AuthType, NoTls, TimeoutSeconds
`ConvertFrom-DirectoryFileTime`	An AD FILETIME (100-ns ticks since 1601) as a DateTime, or $null.	Value
`ConvertFrom-DirectoryGeneralizedTime`	An LDAP generalized time (20260827190803.0Z) as a DateTime, or $null.	Value
`ConvertFrom-DirectoryGuid`	An objectGUID byte array as a GUID string, or $null.	Bytes
`ConvertFrom-DirectorySid`	An objectSid byte array as S-1-5-21-... , or $null.	Bytes
`ConvertTo-DirectoryDnValue`	Escape one RDN value for use in a distinguished name (RFC 4514).	**Value**
`ConvertTo-DirectoryDomainDn`	A DNS domain name as a distinguished name: os7.test -> DC=os7,DC=test.	**DnsDomain**
`ConvertTo-DirectoryFilterValue`	Escape a value for use inside an LDAP search filter (RFC 4515).	**Value**
`ConvertTo-DirectoryInt64`	Internal. An LDAP integer string, parsed invariantly. $null on failure.	Value
`Disconnect-DirectoryServer`	Close a session's connection. Safe to call twice.	**Session**
`Get-DirectoryAccountControl`	Decode userAccountControl into named flags and the two answers an operator actually asks for.	Value
`Get-DirectoryAttributeScalar`	The first value of one attribute, or $null. Never a character.	**Attributes**, **Name**
`Get-DirectoryAttributeValues`	Every value of one attribute, ALWAYS as an array — empty when absent.	**Attributes**, **Name**
`Get-DirectoryErrorMeaning`	Turn an LdapException into a sentence about what is	**Exception**

Command	What it does	Parameters
	wrong.	
`Get-DirectoryIdentityResolution`	Whether the name service can resolve a directory account on this host.	**Name**
`Get-DirectoryKeytabPrincipal`	The principals in a keytab, read with klist -k.	KeytabPath
`Get-DirectoryLdapException`	Internal. Dig the LdapException out of whatever PowerShell wrapped it in.	Exception
`Get-DirectoryRealmConfiguration`	What this host is configured to believe about a realm, read off disk.	SssdConfPath, KeytabPath, Krb5ConfPath
`Get-DirectoryRootDse`	The server's own description of itself: naming contexts, functional levels, the DC's name.	**Session**
`Get-DirectoryTicket`	Get-DirectoryTicket	*none*
`Get-DirectoryWhoAmI`	Ask the SERVER who it thinks this connection is (RFC 4532).	**Session**
`Join-DirectoryRealm`	Join this host to a Kerberos realm with adcli, and write sssd's configuration for it.	**Domain**, UserName, Password, OneTimePassword, ComputerName, OrganizationalUnit, KeytabPath, SssdConfPath, AllowGroup, AllowAllDomainUsers, HomeDirectoryTemplate, LoginShell, UseLdapPassword
`Move-DirectoryEntry`	Move or rename one object.	**Session**, **DistinguishedName**, **NewParentDistinguishedName**, NewName
`New-DirectoryEntry`	Create one object.	**Session**, **DistinguishedName**, **ObjectClass**, Attribute
`New-DirectorySssdConfiguration`	The text of an sssd.conf for a realm. A pure function, so it can be checked without a domain.	**Domain**, AllowGroup, AllowAllDomainUsers, HomeDirectoryTemplate, LoginShell
`New-DirectoryTicket`	Obtain a Kerberos ticket for a principal.	**Principal**, **Password**
`Remove-DirectoryEntry`	Delete one object.	**Session**, **DistinguishedName**
`Remove-DirectoryRealm`	Leave a realm: delete the computer account, the keytab, and the cached	**Domain**, UserName, Password, KeytabPath, SssdConfPath, CacheDirectory

Command	What it does	Parameters
	identities the realm put on this host.	
`Remove-DirectoryTicket`	Destroy the credential cache.	*none*
`Search-Directory`	Run one LDAP search, following the pages to the end.	**Session**, **SearchBase**, **Filter**, Scope, Property, PageSize, SizeLimit
`Set-DirectoryEntry`	Modify attributes of one object.	**Session**, **DistinguishedName**, **Name**, **Value**, Operation
`Set-DirectoryPassword`	Set an account's password through unicodePwd.	**Session**, **DistinguishedName**, **NewPassword**, CurrentPassword
`Split-DirectoryDn`	Split a distinguished name into its components, honouring escapes.	**DistinguishedName**
`Test-DirectoryModule`	Check this module against recorded output from a real directory.	Quiet
`Test-DirectoryTool`	Whether a program this module needs is present. Never throws.	**Name**
`Update-DirectoryRealm`	Renew this host's machine account password and keytab.	Domain, KeytabPath

`OS7`

The product layer: boot environments, releases, backup, management, the directory — everything that is OS/7-specific.

Command	What it does	Parameters
`Add-OS7ADGroupMember`	Add one or more members to a group.	**Identity**, **Member**, Session
`Add-OS7DirectoryTrust`	Trust a domain controller's issuing certificate authority, machine-wide.	**Path**, Name
`Add-OS7RemoteDesktopUser`	Let an account sign in over Remote Desktop.	**Name**
`Disable-OS7ADAccount`	Disable a user or computer account.	**Identity**, Session
`Disable-OS7Backup`	Stop taking scheduled snapshots. Destroys nothing.	*none*
`Disable-OS7RemoteDesktop`	Turn Remote Desktop off.	KeepCredential
`Disable-OS7Remoting`	Stops offering the PowerShell subsystem.	*none*
`Disable-OS7ScheduledTask`	Stops a task's schedule — now, and at the next boot.	**Name**
`Dismount-OS7BackupTarget`	Export a local backup pool and lock its container, so the drive can go.	**Name**
`Enable-OS7ADAccount`	Enable a user or computer account.	**Identity**, Session
`Enable-OS7Backup`	Turn scheduled snapshots on, writing a default policy if there is none.	*none*
`Enable-OS7RemoteDesktop`	Turn Remote Desktop on, so this machine can be reached with mstsc.	AllowFrom, AllowAnySource, Port
`Enable-OS7Remoting`	Makes `Enter-PSSession -HostName` work against this machine.	*none*
`Enable-OS7ScheduledTask`	Makes a task run on its schedule — from now on, and after reboots.	**Name**
`Enter-OS7AdminSession`	Sign in to Active Directory with an administrative account, for this shell.	**Domain**, Credential, Server, UseKerberos, AllowUnencrypted, TimeoutSeconds
`Exit-OS7AdminSession`	Close the directory session and forget it.	Session

Command	What it does	Parameters
`Get-ComputerInfo`	What this machine is, under the property names a Windows script reads.	Property
`Get-OS7AccountLockout`	Whether a wrong password costs anything on this machine, and to whom.	*none*
`Get-OS7ADComputer`	Find computer accounts in Active Directory.	Identity, Filter, SearchBase, Property, Session
`Get-OS7ADDomain`	What the domain controller says about the domain.	Session
`Get-OS7ADDomainController`	The domain controllers a domain publishes in DNS.	**Domain**, First
`Get-OS7ADGroup`	Find groups in Active Directory.	Identity, Filter, SearchBase, Property, Session
`Get-OS7ADGroupMember`	The members of a group.	**Identity**, Recursive, SearchBase, Session
`Get-OS7AdminSession`	The directory session this shell is signed in with, if any.	*none*
`Get-OS7ADObject`	One object by distinguished name, with every attribute it carries.	**DistinguishedName**, Property, Session
`Get-OS7ADOrganizationalUnit`	Find organisational units.	Identity, SearchBase, Session
`Get-OS7ADPrincipalGroupMembership`	The groups an account is in — the inverse of Get-OS7ADGroupMember.	**Identity**, Recursive, ExcludePrimaryGroup, SearchBase, Session
`Get-OS7ADUser`	Find users in Active Directory.	Identity, Filter, SearchBase, Property, Session
`Get-OS7ArcStatus`	Azure Arc: whether the Connected Machine agent is installed, and what it says about itself.	*none*
`Get-OS7BackupCoverage`	Which home directories the policy actually reaches — and which it cannot.	Policy
`Get-OS7BackupPolicy`	What this machine is configured to snapshot, and what it actually has.	ConfigOnly
`Get-OS7BackupStatus`	Whether this machine's data is actually backed up, asked of ZFS.	SkipTargets

Command	What it does	Parameters
`Get-OS7BackupTarget`	Where copies go, and what is actually there.	Name, SkipProbe
`Get-OS7BootEnvironment`	The boot environments on this machine, with what is active and what boots.	Name
`Get-OS7Domain`	Whether this machine is a member of a domain, as configured and as it actually stands.	ProbeAccount
`Get-OS7DomainLogonPolicy`	Which domain groups may sign in to this machine, and which administer it.	SudoersPath
`Get-OS7Endpoint`	The named services OS/7 knows how to test for, out of the data file beside this module.	Cloud, Name
`Get-OS7EntraStatus`	Whether this machine can sign a user in with Entra ID — and if not, why not.	*none*
`Get-OS7FileVersion`	Every version of a file or folder that a snapshot still holds.	**Path**, Newest, DistinctOnly, IncludeCurrent
`Get-OS7Home`	Every home directory on this machine, and whether it has a dataset.	UserName
`Get-OS7InstallLog`	What os7-setup did when this machine was installed.	Path
`Get-OS7IntuneEnrollment`	What can be said about Intune on this machine — and what cannot.	*none*
`Get-OS7KerberosTicket`	The Kerberos tickets this session holds.	*none*
`Get-OS7Log`	The system log, as objects.	Unit, Priority, Since, Until, Tail, Boot, OS7Only
`Get-OS7ManagementStatus`	The three management paths in one answer: can this machine sign users in, be managed, and be inventoried?	SkipNetwork
`Get-OS7NetworkAdapter`	The network adapters this machine has, and what is actually on them.	Name, IncludeLoopback
`Get-OS7NetworkConfiguration`	What this machine is CONFIGURED to do about the	*none*

Command	What it does	Parameters
	network, and what it is ACTUALLY doing — as two separate answers.	
`Get-OS7Release`	What OS/7 releases this machine's channel offers.	Available, Channel, Source
`Get-OS7RemoteDesktop`	Whether this machine can be reached with Remote Desktop, and whether it actually would be.	*none*
`Get-OS7RemoteDesktopCertificate`	The certificate this machine presents to a Remote Desktop client — the one on disk, and the one the daemon is actually serving.	Port
`Get-OS7RemoteDesktopSession`	Who is signed in over Remote Desktop right now.	UserName
`Get-OS7RemoteDesktopUser`	Who may sign in to this machine over Remote Desktop.	*none*
`Get-OS7Remoting`	Whether this machine can be reached with PowerShell — both ways it can be meant.	*none*
`Get-OS7ScheduledTask`	What runs on a schedule on this machine, and whether it actually will.	Name, OS7Only, Detailed
`Get-OS7SecureBoot`	Whether this machine booted with Secure Boot enforcing, and what that does to the running kernel.	Root
`Get-OS7Service`	The services on this machine, and whether they are actually well.	Name, State, OS7Only, Detailed
`Get-OS7Theme`	Reports which desktop appearance this session is actually using.	*none*
`Get-OS7Time`	The clock: local, UTC, the zone, and whether the hardware clock is kept in local time.	*none*
`Get-OS7TimeSynchronization`	Whether this machine's clock is being disciplined, by whom, and whether it is close enough for Entra sign-in to work.	*none*
`Get-OS7Version`	Which OS/7 this is.	Detailed, CheckDrift, Path

Command	What it does	Parameters
`Get-Service`	The services on this machine, in Windows' vocabulary.	Name, **DisplayName**, InputObject, Include, Exclude, DependentServices, RequiredServices
`Join-OS7Domain`	Join this machine to an Active Directory domain.	**Domain**, UserName, Password, OneTimePassword, ComputerName, OrganizationalUnit, AllowGroup, AdministratorGroup, HomeDirectoryTemplate, TargetRoot, SkipServiceRestart, UseLdapPassword
`Mount-OS7BackupTarget`	Unlock and import a local backup pool that is not currently available.	**Name**, Passphrase
`Move-OS7ADObject`	Move an object to another container.	**DistinguishedName**, **TargetPath**, Session
`Move-OS7Home`	Move an existing home directory onto a USERDATA dataset of its own.	**UserName**, RemoveOriginal, SkipContentVerify
`New-OS7ADGroup`	Create a group.	**Name**, **Path**, Scope, DistributionList, Description, Session
`New-OS7ADOrganizationalUnit`	Create an organisational unit.	**Name**, **Path**, Description, Session
`New-OS7ADUser`	Create a user account.	**Name**, **Path**, Password, DisplayName, GivenName, Surname, UserPrincipalName, Mail, Description, Enabled, Session
`New-OS7BackupTarget`	Define where backups go — and, with -CreateOn, build it.	**Name**, **Pool**, **ComputerName**, **CreateOn**, Dataset, **ConfirmDisk**, Passphrase, PoolName, SshKey, Enabled
`New-OS7BootEnvironment`	Clone the running boot environment into a new one, inactive.	Name, From, Release
`New-OS7BootEnvironmentName`	The name of a new boot environment: os7__.	Release, When
`New-OS7KerberosTicket`	Obtain a Kerberos ticket for a principal.	**Principal**, Credential
`New-OS7RemoteDesktopCertificate`	Generate the TLS certificate the RDP listener presents.	Days, Force
`New-OS7Storage`	Create bpool and rpool and lay down the OS/7 dataset hierarchy.	**Root**, **RootDevice**, **BootDevice**, **BootEnvironment**, UserName

Command	What it does	Parameters
`New-Service`	Writes a new service unit and asks systemd whether it loaded.	**Name**, **BinaryPathName**, DisplayName, Description, StartupType, DependsOn, Credential, SecurityDescriptorSddl
`Register-OS7ScheduledTask`	Creates a scheduled task and reports it armed.	**Name**, Command, Execute, Arguments, Daily, Weekly, DayOfWeek, At, OnCalendar, User, Description, Persistent, RandomizedDelay, Disabled, Force
`Remove-OS7ADGroup`	Delete a group, named the way an operator names one.	**Identity**, Session
`Remove-OS7ADGroupMember`	Remove one or more members from a group.	**Identity**, **Member**, Session
`Remove-OS7ADObject`	Delete an object.	**DistinguishedName**, Session
`Remove-OS7ADOrganizationalUnit`	Delete an organisational unit, refusing while anything is still in it.	**Identity**, Session
`Remove-OS7ADUser`	Delete a user account, named the way an operator names one.	**Identity**, Session
`Remove-OS7BackupTarget`	Forget a target. Destroys nothing on it.	**Name**, KeepSyncSnapshots
`Remove-OS7BootEnvironment`	Destroy a boot environment, both halves.	**Name**
`Remove-OS7Domain`	Leave the domain: delete the computer account and remove the credential.	**Domain**, UserName, Password
`Remove-OS7KerberosTicket`	Destroy this session's Kerberos tickets.	*none*
`Remove-OS7RemoteDesktopUser`	Stop an account signing in over Remote Desktop.	**Name**
`Remove-Service`	Removes a service unit this layer could have written.	**Name**, **InputObject**
`Rename-Computer`	Renames this machine.	**NewName**, ComputerName, DomainCredential, LocalCredential, WsmanAuthentication, Force, Restart, PassThru
`Rename-OS7ADObject`	Rename an object, leaving it where it is.	**DistinguishedName**, **NewName**, Session
`Repair-OS7Domain`	Renew this machine's domain credential after it has gone stale.	Domain

Command	What it does	Parameters
`Reset-OS7ADAccountPassword`	Set a user's password.	**Identity**, **NewPassword**, CurrentPassword, MustChangeAtNextLogon, Session
`Restart-Computer`	Restarts this machine — and restarts it, rather than powering it off.	ComputerName, Credential, Force, Wait, Timeout, For, Delay, WsmanAuthentication, Message
`Restart-OS7Service`	Restarts a service and reports what it became.	**Name**
`Restart-Service`	Restarts a service and reports what it became.	**Name**, **DisplayName**, **InputObject**, Include, Exclude, Force, PassThru
`Restore-OS7`	Roll the system back to a previous OS/7 boot environment.	BootEnvironment
`Restore-OS7File`	Copy a file or folder back out of a snapshot.	**Path**, Snapshot, AsOf, Destination, Force
`Resume-Service`	Thaws a frozen service's processes.	**Name**, **DisplayName**, **InputObject**, Include, Exclude, PassThru
`Search-OS7AD`	Run a raw LDAP filter and get the rows back undecorated.	**Filter**, SearchBase, Scope, Property, Session
`Set-OS7AccountLockout`	Turn the account lockout on or off, and set what it costs.	Attempts, LockoutMinutes, ResetMinutes, Disable
`Set-OS7ADAccountExpiration`	Set or clear the date an account stops working.	**Identity**, **DateTime**, **Never**, Session
`Set-OS7ADGroup`	Change a group's description, mail address or display name.	**Identity**, Description, Mail, DisplayName, Attribute, Session
`Set-OS7ADObject`	Set any attribute on any object.	**DistinguishedName**, **Name**, **Value**, Operation, Session
`Set-OS7ADOrganizationalUnit`	Change an organisational unit's description.	**Identity**, Description, Attribute, Session
`Set-OS7ADUser`	Change attributes of a user account.	**Identity**, DisplayName, GivenName, Surname, Mail, Title, Department, Description, Attribute, Session
`Set-OS7BackupPolicy`	Decide what this machine snapshots, and how long it keeps it.	Dataset, Retention, Enabled, Force
`Set-OS7BootEnvironment`	Make a boot environment the one this machine boots.	**Name**, SkipGrubUpdate
`Set-OS7DomainLogonPolicy`	Grant a domain group the right to administer this machine.	**AdministratorGroup**, TargetRoot, SudoersPath

Command	What it does	Parameters
`Set-OS7Mode`	STUB. Sets the OS/7 system mode.	**Mode**
`Set-OS7NetworkAdapter`	Configures an adapter, applies it, checks that it worked — and puts the old configuration back when it did not.	**Name**, **Dhcp**, **Address**, Gateway, Nameserver, SearchDomain, TimeoutSeconds, Force
`Set-OS7RemoteDesktopCertificate`	Use a certificate and key of your own — one issued by an enterprise CA.	**CertPath**, **KeyPath**, InPlace
`Set-OS7RemoteDesktopCredential`	Rotate, or deliberately reveal, the machine-wide Remote Desktop credential.	Rotate, Reveal
`Set-OS7Service`	Whether a service starts at boot.	**Name**, **StartupType**
`Set-OS7Theme`	Switches this user's desktop between OS/7 Classic and stock GNOME.	**Name**
`Set-OS7TimeSynchronization`	Points this machine at NTP servers.	NtpServer, Pool, Exclusive
`Set-OS7TimeZone`	Sets this machine's time zone.	**Id**
`Set-OS7UpdateChannel`	Point this machine at an OS/7 repository, and choose its channel.	Channel, Uri, Disable
`Set-Service`	Changes whether a service starts at boot, or starts and stops it.	**Name**, **InputObject**, DisplayName, Description, StartupType, Status, Credential, SecurityDescriptorSddl, Force, PassThru
`Set-TimeZone`	Sets this machine's time zone.	**Id**, **Name**, **InputObject**, PassThru
`Start-OS7Backup`	Snapshot now, replicate now, or both — without waiting for the timer.	Snapshot, Replicate, Target
`Start-OS7BackupReplication`	Send this machine's snapshots to its targets, and check they arrived.	Target, Force
`Start-OS7ScheduledTask`	Runs a task now, waits for it, and reports what happened.	**Name**
`Start-OS7Service`	Starts a service and reports what it became.	**Name**
`Start-Service`	Starts a service and reports what it became.	**Name**, **DisplayName**, **InputObject**, Include, Exclude, PassThru

Command	What it does	Parameters
`Stop-Computer`	Powers this machine off.	ComputerName, Credential, Force, WsmanAuthentication, Delay, Message
`Stop-OS7RemoteDesktopSession`	Sign a Remote Desktop user out, ending everything they had open.	**Id**, Force
`Stop-OS7Service`	Stops a service and reports what it became.	**Name**
`Stop-Service`	Stops a service and reports what it became.	**Name**, **DisplayName**, **InputObject**, Include, Exclude, Force, NoWait, PassThru
`Suspend-Service`	Freezes a service's processes — systemd's nearest thing to a paused Windows service.	**Name**, **DisplayName**, **InputObject**, Include, Exclude, PassThru
`Sync-OS7Time`	Asks chrony to correct the clock now, and reports what it did.	SettleSeconds
`Test-OS7Backup`	Check the backup layer's decisions, offline or against this machine.	Live
`Test-OS7BackupTarget`	Is this target reachable, healthy, and holding what it should?	Name
`Test-OS7Directory`	Whether this machine can reach and use a domain controller, and which part is broken when it cannot.	**Domain**, Server, Credential, AllowUnencrypted
`Test-OS7Domain`	Whether the domain membership actually works, and which part does not.	Domain, ProbeAccount
`Test-OS7Network`	Whether this machine can actually reach the network, its resolver and the services OS/7 exists to reach.	Endpoint, Cloud, TimeoutSeconds
`Test-OS7RemoteDesktop`	Check the Remote Desktop configuration on this machine, one question at a time.	*none*
`Test-OS7Update`	Check the update train's decisions. No ZFS, no repository, no VM.	*none*
`Unlock-OS7Account`	Clear an account's failed sign-in count, so it can be used again.	**Name**

Command	What it does	Parameters
`Unlock-OS7ADAccount`	Unlock an account that lockout policy has locked.	**Identity**, Session
`Unregister-OS7ScheduledTask`	Removes a task Register-OS7ScheduledTask created — and only such a task.	**Name**
`Update-OS7`	Apply the next curated OS/7 release into a new boot environment.	Version, Channel, Source, Stage, Reboot, Keep, AllowDevelopment, Force

Appendix B Files and paths

Where things live on an OS/7 machine — for when a cmdlet is not to hand, or you want to see what one rests on.

Identity and release

Path	What is in it
`/usr/lib/os7/release.json`	the complete release description: version, channel, archive snapshot, components. `Get-OS7Version` reads it, `os7-setup` reads it, and the boot environment's name is built from it
`/usr/lib/os7/product`	the product name for every surface a person sees. A file of its own, so the branding depends on no `os-release` field
`/etc/os-release`	`PRETTY_NAME` is branded; `NAME`, `ID`, `VERSION` and `VERSION_ID` deliberately are not
`/etc/issue`	what stands at the console before sign-in
`/usr/lib/os7/migrations/`	the per-release migrations, as `<version>/<chroot|firstboot>/NN-name`

Storage

Path	What it is
`rpool/ROOT/os7_<version>_<time>`	the boot environment — `/`
`bpool/BOOT/os7_<version>_<time>`	its `/boot`
`rpool/USERDATA/<user>_<uuid>`	a home directory, outside the boot environment
`rpool/DATA/…`	state that survives a rollback: `log`, `spool`, `tmp`, `srv`, `lib/<service>`, `snapd`
`/dev/disk/by-partlabel/os7-luks`	the encrypted container `rpool` lives in

Boot

Path	For
`/etc/grub.d/09_os7-boot-environments`	OS/7 generates its own boot menu here, one entry per boot environment
`/boot/efi/…/grub.cfg`	the file on the EFI partition that decides whose menu GRUB reads
`<boot environment>/boot/grub/grubenv`	`saved_entry` — which entry inside it starts
`/proc/cmdline`	the kernel command line. `boot=zfs` must be on it

Configuration

Path	For
`/etc/netplan/*.yaml`	the network configuration. `Get-OS7NetworkConfiguration` merges it and names each statement's origin
`/etc/chrony/sources.d/*.sources`	the NTP sources. `Set-OS7TimeSynchronization` writes here, not into `chrony.conf`
`/etc/localtime`	the symlink that decides the time zone
`/etc/os7/backup.json`	the backup policy. Authoritative; the sanoid configuration is generated from it
`/etc/apt/sources.list.d/os7-release.*`	the OS/7 repository. Shipped deliberately disabled

Logs

Path	What
`/var/log/os7-setup/install.log`	the installation record, with each step's self-proof. Mode `0600`, **inside** the boot environment. `Get-OS7InstallLog`
the systemd journal	`Get-OS7Log`. It lives under `/var/log`, so **outside** the boot environment — it survives the rollback of the update it explains

Programs and modules

Path	What
`/usr/lib/os7-setup/os7-setup`	the installer. `--self-test`, `--version`, `--unattend`, `--dry-run`
`/usr/local/share/powershell/Modules/`	the six modules: `OS7`, `Zfs`, `Net`, `Time`, `Systemd`, `Directory`
`/usr/share/consolefonts/os7-console-16x32.psf.gz`	the installed console's font (Cascadia Mono)
`/usr/share/consolefonts/os7-fixedsys-16x32.psf.gz`	the installer's font (Fixedsys)
`/etc/profile.d/95-os7-powershell.sh`	the hand-off of the login shell to PowerShell

Services

Unit	For
`os7-update-check.timer` / `.service`	the unattended update check
`os7-backup-replicate.service`	the replication run
`os7-migrations-firstboot.service`	the first-boot migrations after an update
`zfs-import-cache`, `zfs-mount`, `zfs-zed`	the ZFS chain at boot
`chrony.service`	the clock
`authd.service`	sign-in with Entra accounts

Appendix C For Windows administrators

What you do on Windows, and how it is done on OS/7. The table is a map, not an equivalence — where the concepts differ, it says so.

The system and its version

Windows	OS/7
`winver`, `Get-ComputerInfo`	`Get-OS7Version`, `Get-OS7Version -Detailed`
`systeminfo`	`Get-OS7Version -Detailed`, `Get-OS7ManagementStatus`
Control Panel → System	`Get-OS7Version`, `Get-OS7Home`, `Get-Zpool`

Services

Windows	OS/7
`Get-Service`	`Get-OS7Service`
`Get-Service | Where Status -eq Running`	`Get-OS7Service -Detailed | Where { $_.Healthy -eq $false }`
`Start-Service`, `Stop-Service`, `Restart-Service`	`Start-OS7Service`, `Stop-OS7Service`, `Restart-OS7Service`
`Set-Service -StartupType`	`Set-OS7Service -StartupType`
`services.msc`	`Get-OS7Service -OS7Only -Detailed`

> **The left-hand column works here too.** `Get-Service`, `Start-Service`, `Stop-Service`, `Restart-Service` and `Set-Service` exist on OS/7 with Windows' parameters and Windows' columns (`Status`, `Name`, `StartType`, `DisplayName`) — PowerShell itself does not ship them on Linux. The right-hand column stays the surface this manual teaches: its parameters are systemd's, and `Healthy` answers a question `Status` cannot — a unit in a restart loop reports `Running`.
>
> Two differences matter if you use the Windows names. `Set-Service -StartupType Disabled` **masks** the unit, because "cannot be started" is exactly what that means to systemd; `-StartupType Manual` undoes it. And `Suspend-Service` freezes the processes rather than asking the service to pause — systemd does not ask.

Scheduled tasks

Windows	OS/7
`Get-ScheduledTask`, `taskschd.msc`	`Get-OS7ScheduledTask`
`Get-ScheduledTaskInfo`	`Get-OS7ScheduledTask <name>` — `NextRun`, `LastRun`, `LastResult`
`Enable-ScheduledTask`, `Disable-ScheduledTask`	`Enable-OS7ScheduledTask`, `Disable-OS7ScheduledTask`
`Start-ScheduledTask`	`Start-OS7ScheduledTask`
`Register-ScheduledTask`	`Register-OS7ScheduledTask`
`Unregister-ScheduledTask`	`Unregister-OS7ScheduledTask`
`schtasks /create`	`Register-OS7ScheduledTask -Daily -At 03:00 -Command '…'`

Event log

Windows	OS/7
`Get-WinEvent`, `eventvwr`	`Get-OS7Log`
`Get-WinEvent -MaxEvents 50`	`Get-OS7Log -Tail 50`
filter by source	`Get-OS7Log -Unit ssh.service`
filter by level	`Get-OS7Log -Priority Error`
events since the last boot	`Get-OS7Log -Boot 0`, before that `-Boot -1`

Network

Windows	OS/7
`Get-NetAdapter`	`Get-OS7NetworkAdapter`
`Get-NetIPConfiguration`	`Get-OS7NetworkConfiguration`
`New-NetIPAddress`, `Set-DnsClientServerAddress`	`Set-OS7NetworkAdapter -Address … -Nameserver …`
turn on DHCP	`Set-OS7NetworkAdapter -Dhcp`
`Test-NetConnection`	`Test-OS7Network`, or PowerShell's own `Test-Connection`
`ipconfig /all`	`Get-OS7NetworkAdapter | Format-List`

Time

Windows	OS/7
`w32tm /query /status`	`Get-OS7TimeSynchronization`
`w32tm /resync`	`Sync-OS7Time`
`Set-TimeZone`	`Set-OS7TimeZone -Id Europe/London`
`Get-Date`	`Get-Date` (unchanged); `Get-OS7Time` for the full answer

Disks

Windows	OS/7
Disk Management, `Get-Disk`, `Get-Volume`	`Get-Zpool`, `Get-ZfsDataset`
`Get-Volume | Select SizeRemaining`	`Get-ZfsSpace`
growing a partition	does not arise — datasets share the pool's free space
`chkdsk`	`Start-ZpoolScrub` (verifies all data against its checksums)
Volume Shadow Copies	ZFS snapshots: `New-ZfsSnapshot`, `Get-ZfsSnapshot`
Previous Versions of a file	`Get-OS7FileVersion`, `Restore-OS7File`

Updates and recovery

Windows	OS/7
Windows Update	`Get-OS7Release`, `Update-OS7`
WSUS ring / servicing ring	`Set-OS7UpdateChannel -Channel stable|preview|development`
uninstall an update	`Restore-OS7` — goes back to the previous boot environment
System Restore point	`New-OS7BootEnvironment`
managing restore points	`Get-` / `Remove-OS7BootEnvironment`
Safe Mode	pick an older boot environment in the boot menu

> The difference from Windows is fundamental: an update on OS/7 does **not** change the running system. It fills a copy and switches afterwards. There is therefore no such state as "half updated".

Directory

Windows (RSAT / ActiveDirectory module)	OS/7
`Get-ADUser`	`Get-OS7ADUser`
`New-ADUser`	`New-OS7ADUser`
`Set-ADUser`	`Set-OS7ADUser`
`Get-ADGroup`, `Get-ADGroupMember`	`Get-OS7ADGroup`, `Get-OS7ADGroupMember`
`Add-ADGroupMember`	`Add-OS7ADGroupMember`
`Set-ADAccountPassword`	`Reset-OS7ADAccountPassword`
`Unlock-ADAccount`, `Enable-` / `Disable-ADAccount`	`Unlock-OS7ADAccount`, `Enable-` / `Disable-OS7ADAccount`
`Get-ADComputer`, `Get-ADOrganizationalUnit`	`Get-OS7ADComputer`, `Get-OS7ADOrganizationalUnit`
`Get-ADObject -LDAPFilter`	`Search-OS7AD -Filter`
`Add-Computer -DomainName`	`Join-OS7Domain`
`Test-ComputerSecureChannel`	`Test-OS7Domain`
`Reset-ComputerMachinePassword`	`Repair-OS7Domain`
`klist`, `klist purge`	`Get-OS7KerberosTicket`, `Remove-OS7KerberosTicket`

Unlike on Windows, directory administration needs **no domain membership**: `Enter-OS7AdminSession` opens a session from any OS/7 machine.

Remoting

Windows	OS/7
`Enter-PSSession -ComputerName` (WinRM)	`Enter-PSSession -HostName` (SSH)
`Invoke-Command -ComputerName`	`Invoke-Command -HostName`
`Enable-PSRemoting`	`Enable-OS7Remoting`
RDP	SSH; on the desktop, the usual Linux tools as well

> There is no WinRM here. `-ComputerName` exists as a parameter and answers "no supported WSMan client library was found". SSH-based remoting is PowerShell's own mechanism and works from Windows too.

What stayed the same

These cmdlets work unchanged on OS/7 and were therefore deliberately **not** rebuilt:

`Get-Process` · `Stop-Process` · `Get-FileHash` · `Get-Date` · `Test-Connection` · `Get-Credential` · `Get-ChildItem` · `Copy-Item` · `Select-String` · `ConvertTo-Json` · `Invoke-RestMethod` · `Get-Content` · `Start-Job`

`Restart-Computer` and `Stop-Computer` were on that list until 2026-09-09 and are now on the next one: **OS/7 supplies them itself.** PowerShell's own version on Linux runs `/usr/sbin/shutdown` with **no arguments at all**, and on Ubuntu that path is a symlink to `systemctl`, whose default action without a flag is **power off**. So `Restart-Computer` powered the machine off and reported success. The OS/7 version says `systemctl reboot`, and it takes the `-Force` a copied script carries — PowerShell's own has no parameters at all on Linux.

Windows names that exist here

Fourteen cmdlets from `Microsoft.PowerShell.Management` that PowerShell does not ship on Linux (measured: 15 of the 62 documented ones are absent) are supplied by OS/7 itself, with Windows' parameters:

Name	Note
`Get-Service`	`Status` is `Running` / `Stopped` / `Paused`, `StartType` is `Automatic` / `Manual` / `Disabled`. systemd's own words sit beside them: `SystemdActiveState`, `SystemdSubState`, `SystemdFreezerState` — and `Healthy`.
`Start-` / `Stop-` / `Restart-Service`	Each reads the unit back afterwards. `systemctl start` reports success as soon as the *job* is accepted.
`Suspend-` / `Resume-Service`	systemd's freezer. It freezes every process in the unit without asking. A frozen unit still reports `ActiveState=active`, which is why `Paused` comes from the freezer.
`Set-Service`	`-StartupType` and `-Status`. `Disabled` masks.
`New-` / `Remove-Service`	Writes or removes a `.service` unit in `/etc/systemd/system` and asks systemd whether it loaded. It removes nothing a package brought.
`Set-TimeZone`	The half that was missing beside `Get-TimeZone`. Takes IANA names **and Windows ids**: `-Id 'W. Europe Standard Time'` becomes `Europe/Berlin`.
`Get-ComputerInfo`	What this machine is, under Windows' property names.
`Rename-Computer`	Changes the static name, the kernel name and the `127.0.1.1` line in `/etc/hosts` together. **Refused on a domain-joined machine** — the keytab carries the old name.
`Restart-` / `Stop-Computer`	See above.

> Anything a cmdlet here cannot do, it refuses **by name** and says where to go: `Get-Service -DependentServices` explains that systemd keeps dependencies as a graph and that `systemctl list-dependencies --reverse` answers the question. A parameter ignored in silence would be worse than an error.

And so does the whole language: pipelines, `Where-Object`, `ForEach-Object`, the format cmdlets, `Export-Csv`, error handling, scripts and modules.

What is not here

Windows	Why not
Group Policy / the GPO editor	there is no GPO engine for Linux
`repadmin`, `dcdiag`, `netdom`	RPC/DCOM; no cross-platform client
DNS and DHCP server management	the same cause
`[ADSI]`	`System.DirectoryServices` loads on Linux and then throws "not supported on this platform". A Windows script built on it does not port by copying
WinRM	see above